# 1. About this Open Learning Module

## Target Audience and Prerequisites

This Open Learning module is aimed at new recruits in Great Britain into the role of Signalling Technician on either the installation or maintenance sides. It can also be used to update or refresh the knowledge of experienced technicians who are undergoing re-certification of their competence.

Apart from a basic Secondary education, an ability to understand moderately complex English language and the presentation of information in the form of diagrams, plus a reasonable curiosity and a willingness to learn, the only requirement is that you should have successfully completed study of the associated Open Learning Module “Introduction to Railway Signalling”, or have acquired equivalent basic knowledge about the purpose and implementation of railway signalling.

## Objectives

This module is intended to provide an introduction to the various types of electromechanical relays used in railway signalling, and the principles of their operation. It covers

- relay component identification
- operation of a DC neutral relay
- the main relay types
- the relay room
- types of faults

## What you will require

Apart from this Open Learning manual the only other material you will require is the Signalling Installation Handbook (GS/IH).

## The Results expected from this Module

When you have completed this module, the assessment and been awarded your certificate, you should

- be able to identify the components of relays
- understand the principles of operation of relays
- be able to correctly recognise the main relay types
- have an appreciation of the relay rooms and locations where relays are installed
- be able to locate and identify the function of specific relays

## An Introduction to Jargon

You may have wondered about the word **jargon**. When talking to a friend you often talk in a way that you both understand but which a stranger may find very difficult to understand. For example, you might say “the Reds were terrific on Saturday”, meaning the local football team had done well.

Throughout this module you will find yourself introduced to a variety of jargon used on the railways. This will help you to understand not only training modules but also some of the language which signal persons, signalling engineers, signalling technicians and others use when speaking or writing about signalling equipment and systems. You will find the same jargon is used in the many written rules, specifications and drawings you will come across in the course of your work.
The Glossary of Terms (see Section 9) introduces you to some of the variations in the jargon.

## A Safety Warning

This Open Learning module is designed to arouse your curiosity about railway signalling. However, please remember that, until you have undergone proper training and been certified as competent in Personal Track Safety (PTS) and you have been authorised by your supervisor, you are forbidden to go onto any part of the railway that is not open to passengers, unless you are properly supervised and it is an official part of your training.

Study and remember what it says in your Rule Book and the Track Safety Handbook. There is no requirement for you to go onto the railway to successfully complete this course of study.

Of course, your supervisor or appointed Mentor may use their discretion and properly arrange for you to go onto the railway to look at practical examples of the signals, associated equipment and junctions described in this manual and which you have not already had an opportunity to observe in the course of your work and other training.

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# 2 Relay Component Identification

## 2.1 Introduction

Within the railway signalling system situations regularly arise where

- one electrical circuit is needed to control another
- the two circuits must not have an electrical connection between them

In a case such as this, a remotely controlled switch is required and the simplest method which can be used to achieve this operation is to employ a **relay**.

*Control of Railway Signalling Circuitry*

Relays are central to most inter-locking systems. They form an important part of the system being essential devices in the control of railway signalling circuitry including

- track circuits
- signals
- points

In this section you will look at the **component parts** which make up a relay. In Section Two you will examine what roles the component parts play in the **operation** of the relay.

At this stage take your time to look at the components which make up the relay as this familiarisation will help you when you go on to look at the actual operation.

A relay is an **electrically operated switch** for opening and closing circuits.

*Three Principal Components*

The most common forms of relay are magnetically operated and have three principal components

- coil
- armature
- contacts

These three component parts are adapted to suit the specific function for which the relay is employed.

*930 Series Miniature Relays*

The physical design of these components can vary considerably. In this section the British Rail Specification 930 DC Neutral Relay will be examined. The Specification 930 relay is one of a series of so-called **miniature relays** which are known collectively as the **930 series**.

In addition to the specification 930 relay other relays in this series will be examined in later sections of this module. These relays are generally designed to operate with a coil supply voltage of 50 V DC but versions to operate with 24 V DC and other voltages are available.

Look carefully at Figure 1.

*Construction of a Specification 930 Relay*

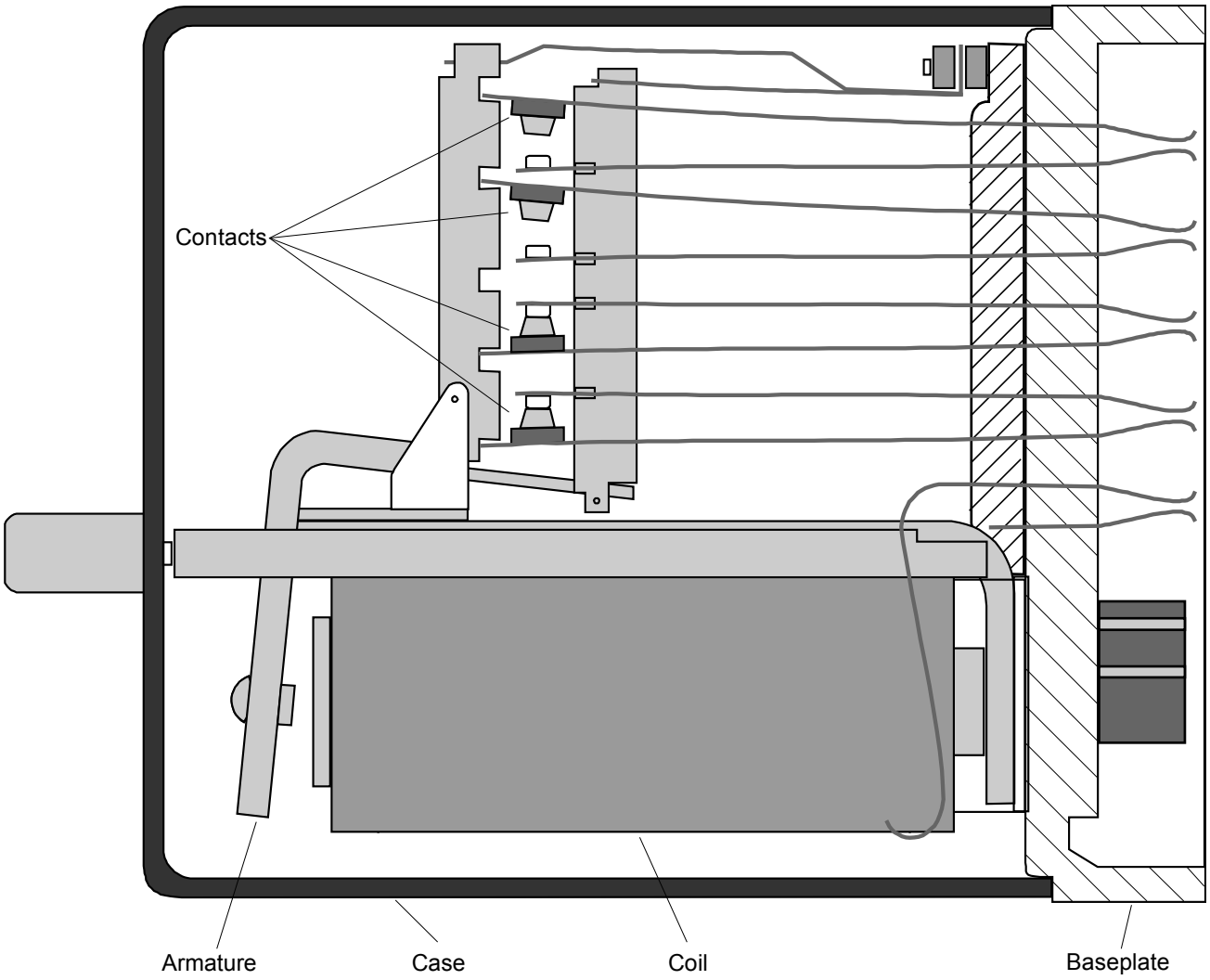



Figure 1

*Part Section of a Relay*

Figure 1 shows the construction of a complete specification 930 DC neutral relay.

You will now look at the components of this relay in greater detail. Figure 2 provides a section of Figure 1 in order to make the component examination easier.

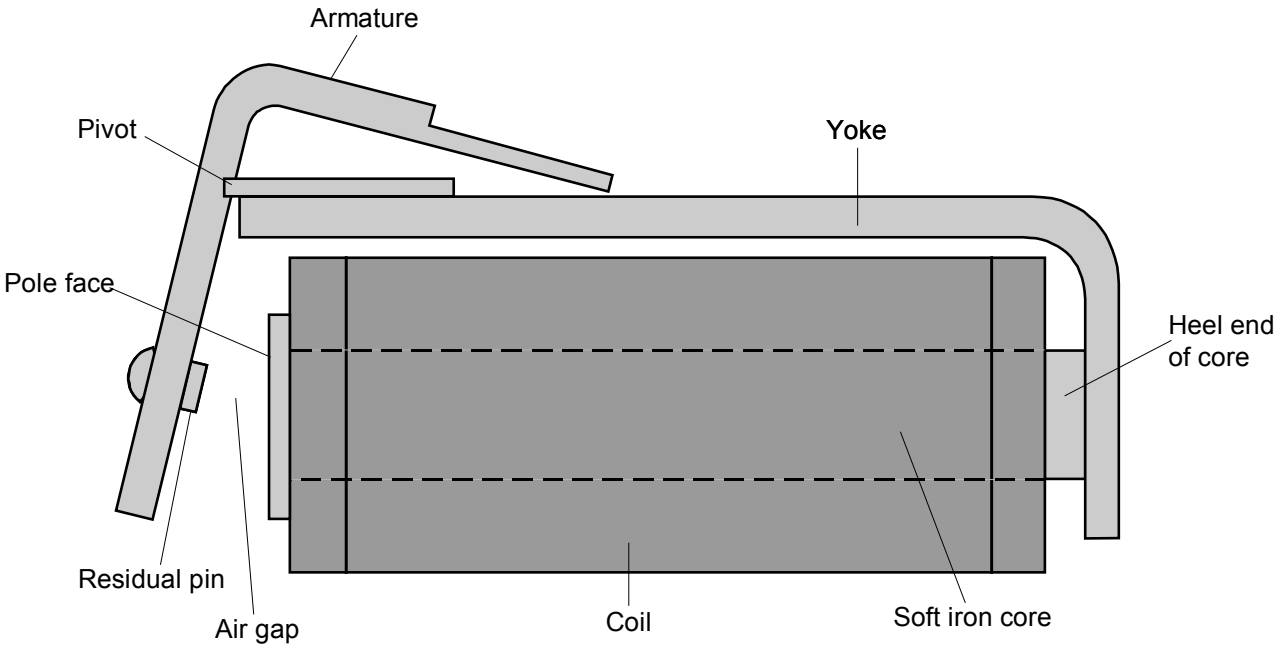



Figure 2

## 2.2 Coil

The coil consists of a very long length of enamelled, thin copper wire wound on a **soft iron core**. To keep it in place, the coil is wound on a former and bound with a protective tape.

## 2.3 Yoke

The yoke and **heel end** of the soft iron core are joined together to form a magnetic circuit. Soft iron is used because it is permeable to the magnetic flux generated when current flows in the coil, but retains little of that flux when the current is cut off.

## 2.4 Armature

The soft iron armature is the moving part of the mechanism. It is attached to the yoke at the **pivot**. When current flows in the coil, the resulting magnetic flux attracts the armature to the **pole face**.

## 2.5 Residual Pin

The residual pin is a small stud of non-magnetic material which is fitted to ensure a definite minimum **air gap** between the armature and the core when the relay is **energised**. The term energised will be explained in a later section. The air gap will prevent any residual flux keeping the armature stuck to the pole face when the supply to the coil is disconnected. This will become clearer when you look at the operation of the relay.

## 2.6 Contacts

The relay contains a number of contacts which are given the following descriptions

*Front and Back Contacts*

- front contact
- back contact

The front and back contacts are to be considered in greater detail later in the text.

*Drive Bar*

The contacts are shown in Figure 3. They are mounted above the relay coil and are operated by a **drive bar** connected to the armature.

Contacts are made from specific materials in order to

- avoid contacts welding together
- provide current carrying capacity
- provide low contact resistance

*Welding of the Contacts*

In signalling safety relays it is extremely important to avoid **welding** of the contacts. Consequently, it is usual to utilise the following

- silver impregnated graphite for fixed contacts
- silver for moving contacts

*Relay Contact Assembly*

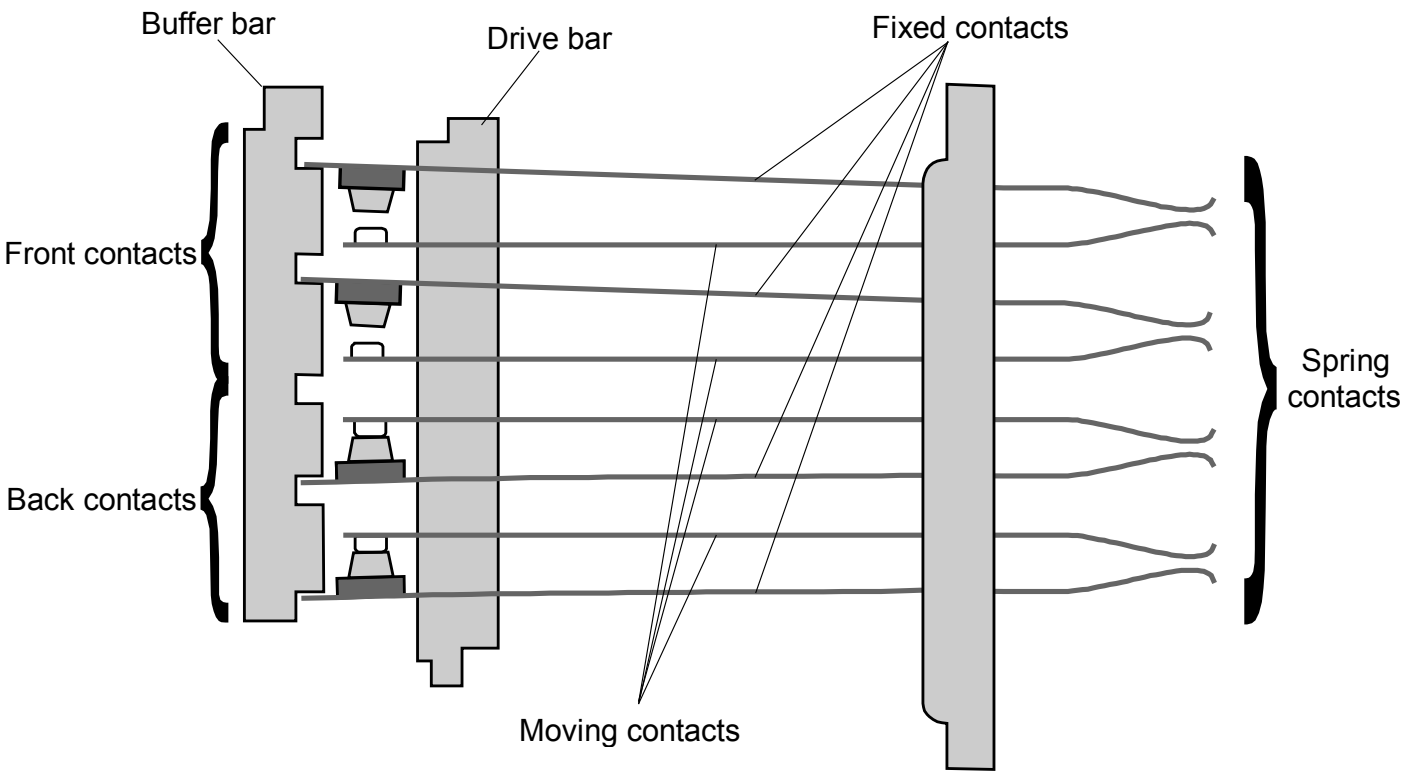



Figure 3

## 2.7 Buffer Bar

The buffer bar limits the travel of the contacts, it also holds the fixed contacts in position in order to maintain correct contact pressure.

## 2.8 Spring Contacts

The spring contacts are the electrical terminations which provide for external connections to the relay. The association of the spring contacts with the plug board is explained later.

Figure 3 and Figure 4 illustrate the spring contacts.

## 2.9 Base Plate

The components of the relay are assembled on the base plate.

Figure 1 and Figure 4 provide an illustration of the base plate.

## 2.10 Relay Casing

The relay is enclosed in a clear plastic case to provide protection from

*Relay Damage*

- dirt
- moisture
- other environmental damage
- physical damage

*2.11*

## Seal

The seal is normally made from either

- wax
- wire secured with lead

*Retaining Screws*

The seal secures the **retaining screws**. The seals must not be broken. If a relay has a broken seal it **must not** be used.

*Typical Front and Rear Views of a 930 Series Relay*

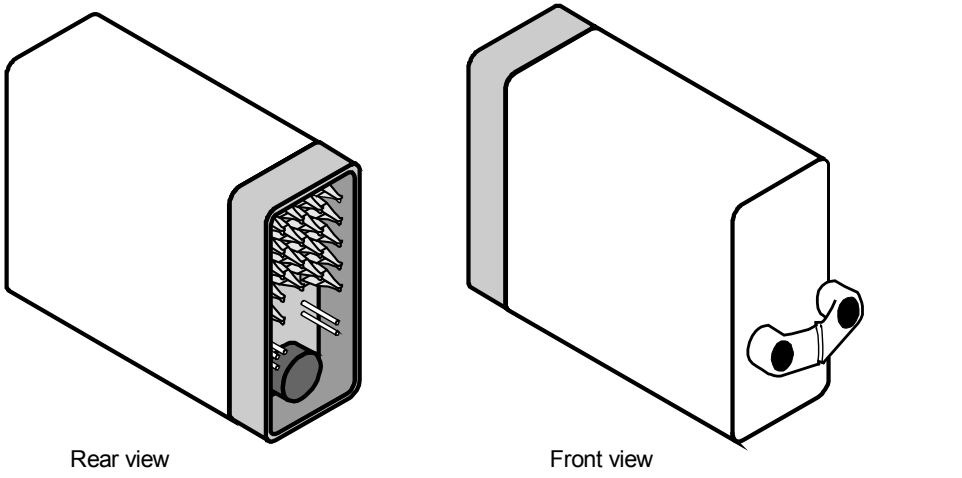



Figure 4

## 2.12 Configuration Pin Coding

Each 930 series relay is configured by a number of **pins** protruding from it. Individual versions of relays have a specific pin pattern according to a pre-set code and the associated plug boards have corresponding holes. This means that it is impossible to plug a different type of relay into a particular plug board. The number of pins used is either five or six according to the type of relay.

Figure 5 provides some examples of pin coding configuration extracted from The Signalling Installation Handbook Section 2C10.

*Pin Coding Configuration*

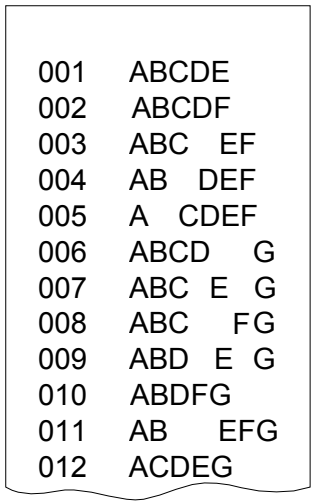



Figure 5 006

It is worth looking at the Section 2C10 of the Signalling Installation Handbook in order to gain an understanding of how the pin configuration relates to the code numbers.

## 2.13

# Plug board

Look carefully at Figure 6.

*Relay*
*Plug board*

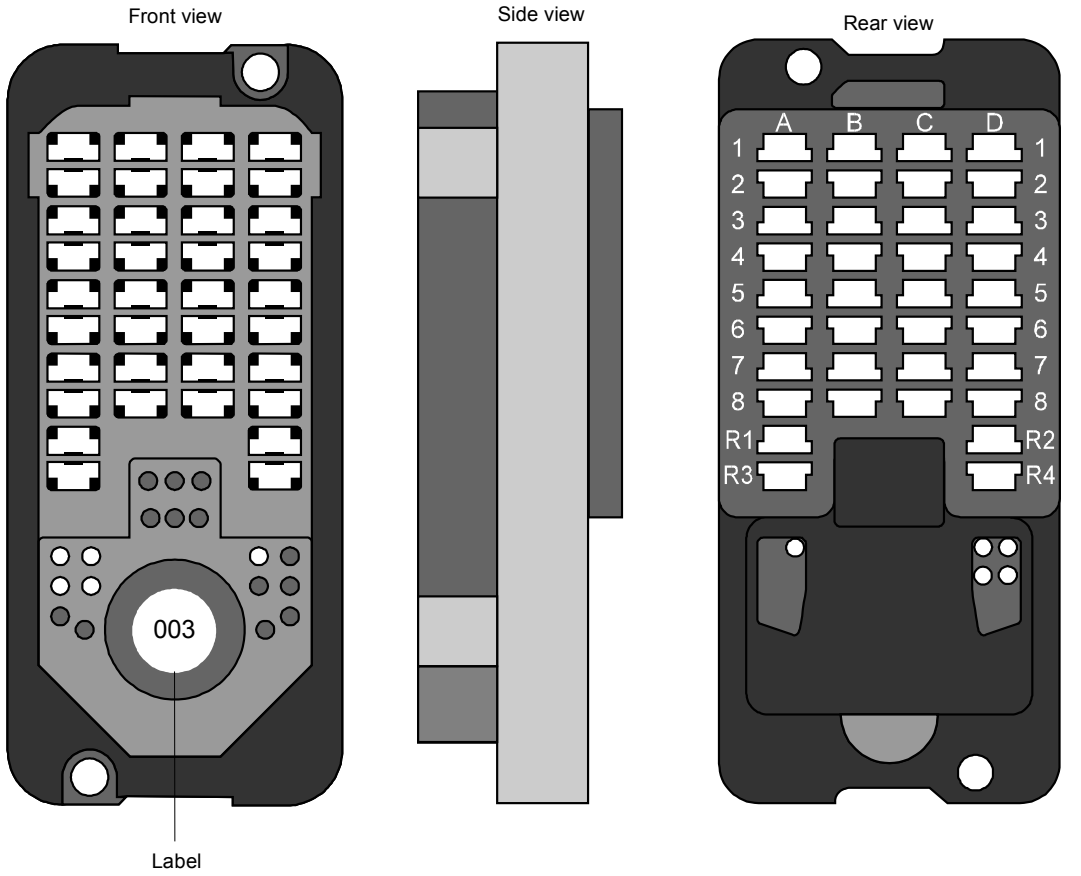



Figure 6

Figure 6 provides an example of the plug board. The plug board is designed to receive all versions of the 930 series relays. Plug boards drawn from general stock do not have configuration holes drilled in them. The hole positions are marked by dimples, which act as drilling guides to allow drilling as required.

*Plug board Drilling*

When a plug board has been drilled a label should be attached to the front showing the pin code number.

Note that the two fixing holes positioned at the top and bottom of the plug board provide orientation.

## 2.14 Retaining Clip

*Retaining Clip Tool*

The retaining clip secures the plug-in relays to the plug board. It is important to fit the retaining clip using the correct tool.

Screwdrivers, pliers or other non approved tools should not be used as levers. An example of a retaining clip can be seen in Figure 8.

## 2.15 Relay Labelling and Marking

As there are a number of different types of relays it is important that all relays should have either

- a label or
- some form of marking

*Relay Label Information*

The information given on the relay should include all of the following

- manufacturer's name
- manufacturer's reference or serial number
- BR specification number
- pin code
- rated coil voltage
- contact arrangement
- date tested

*Other Label Information*

The information may also include

- BR catalogue number
- date serviced

Each relay coil may also show the resistance value marked on the coil body.

*Selecting a Relay*

When selecting a relay make sure you look for the

- pin code
- rated voltage
- contact arrangement
- BR specification number

## 2.16 Spades

External connections to the relay numbered on the plug board are made by spades (spade connectors) which fit into and are retained by the plug board. Each spade can accommodate only one or two wires. To correctly make contact with its associated relay spring contact, each spade must be correctly located in its slot in the plug board.

An example of a spade is illustrated in Figure 7, which shows it both enlarged (to expose the detail) and actual size. Also illustrated is the manner of attaching the wires by crimping.

*Spade*

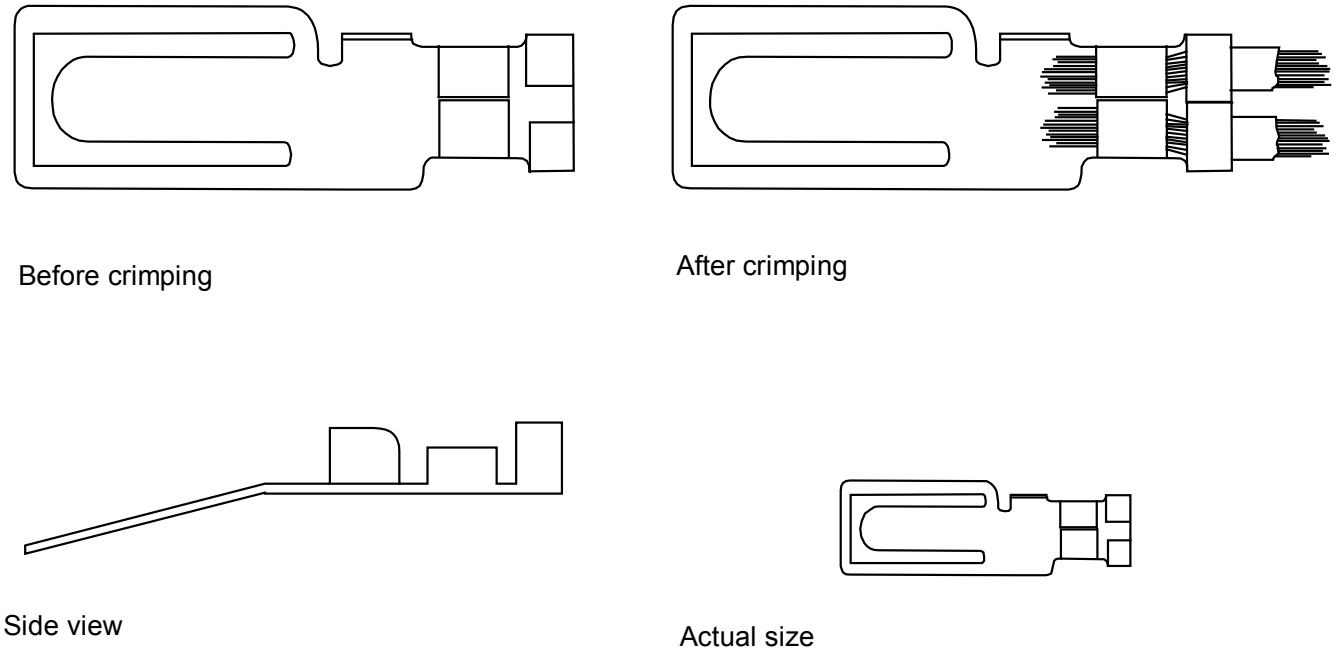



Figure 7

## 2.17 Beads

Beads identify the wiring to the relay plug board spades. An example of this can be seen in Figure 8.

*Plug board, Wiring and Labelling*

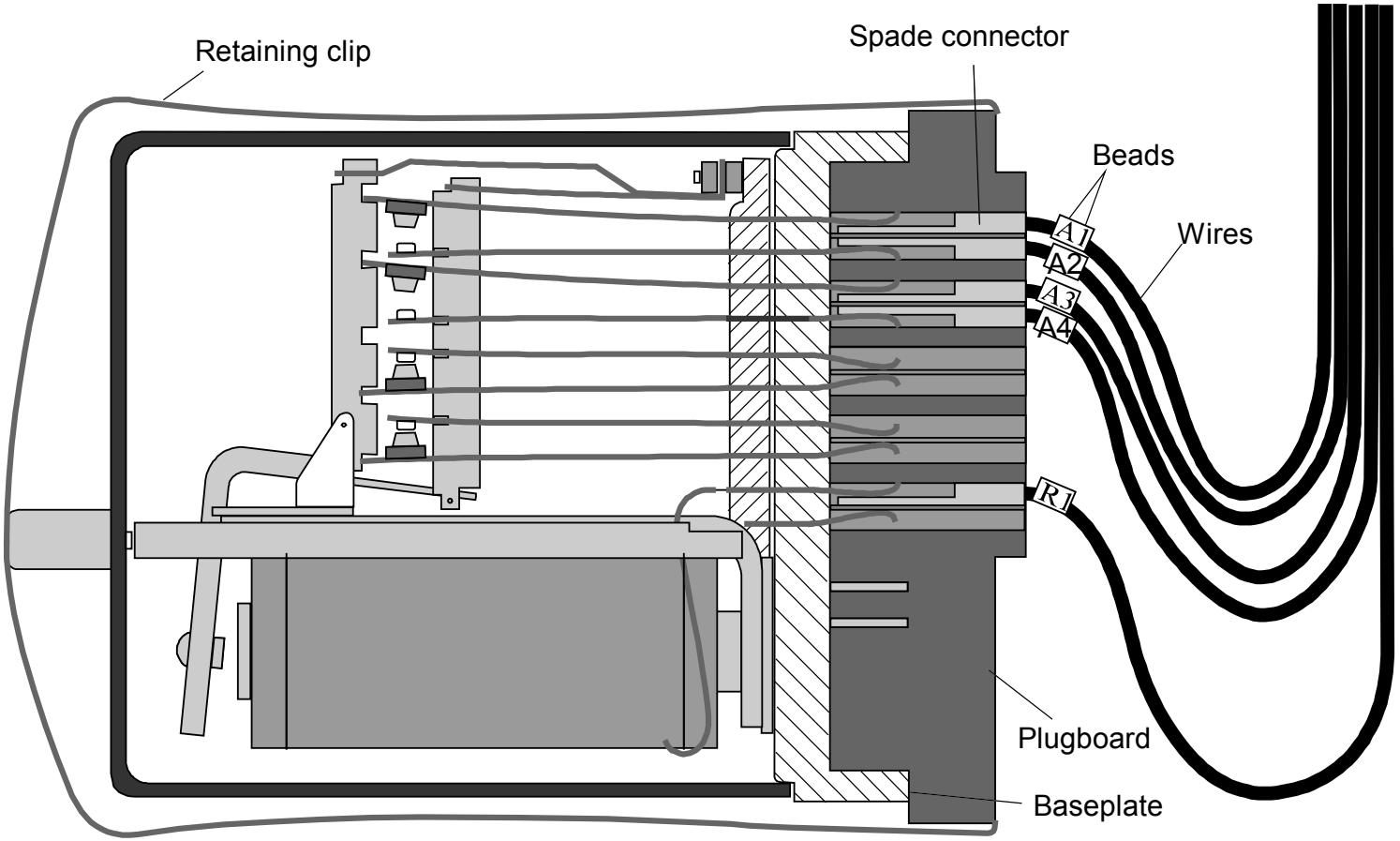



Figure 8

## 2.18 Interconnections and Contact Stacks

As already described, each spade connector can accept only one or two wires. When more than two spades have to be joined together, it is necessary for the designer to arrange the interconnections in a “daisy chain” fashion, organised so that the amount of wire used is kept to a reasonable minimum.

Figure 8 (and other illustrations which show a side view of a relay ) displays only one stack of the contacts. In practice, each relay plug board accommodates the connections to four stacks of contacts and there can be up to 14 such plug boards in a row. Consequently, most of the interconnecting wires are routed more horizontally than vertically and careful routing is necessary to avoid the creation of congested masses of wires.

## 2.19 Section Summary

Section Two looked at the component parts of a relay.

Now complete Self-Assessment Question 1 before proceeding to Section Three.

### Self-Assessment Question 1

Correctly identify the component parts shown on Figure 9.

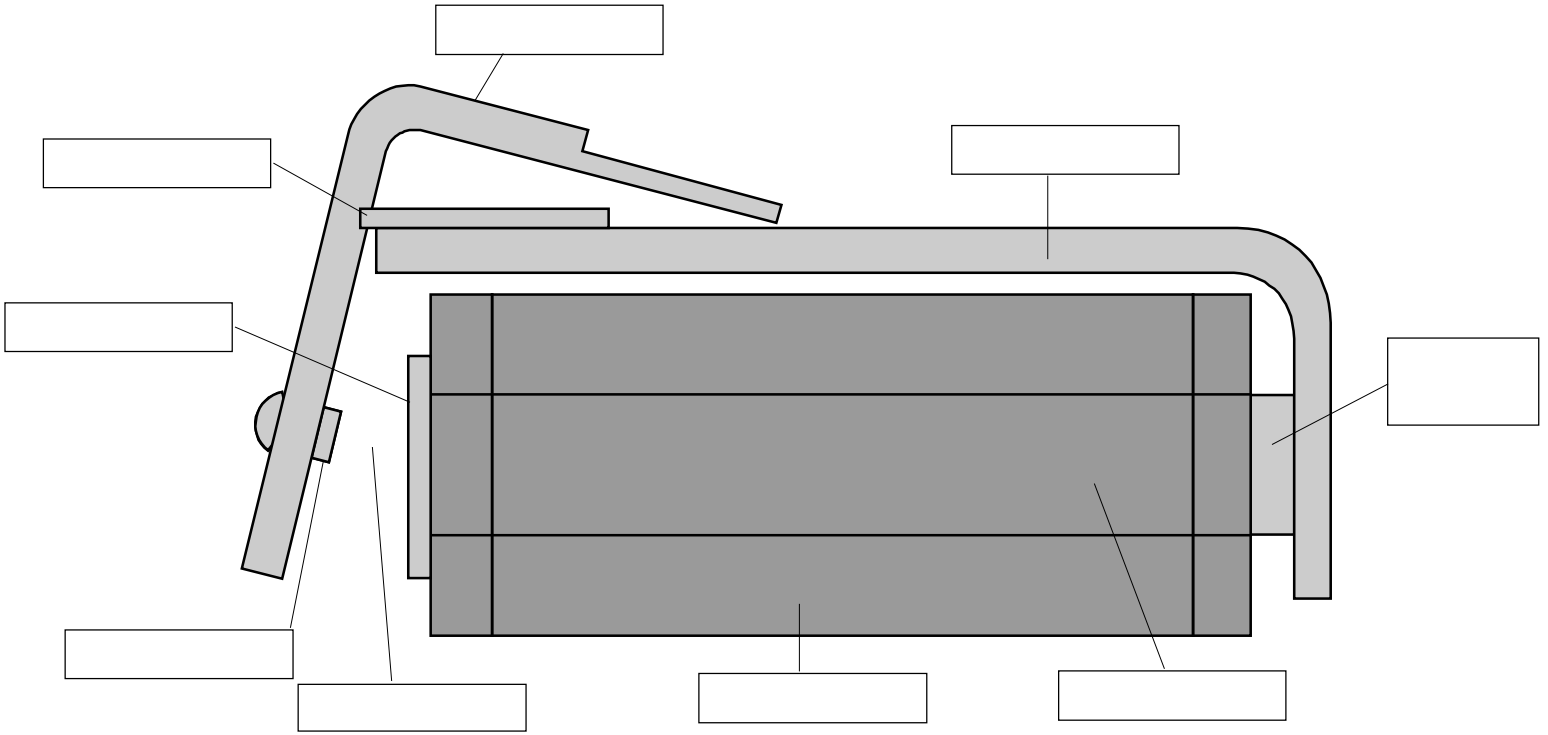


Figure 9

009

Now check your answers with those given in the Answers Section. Before moving on to Section Three make sure that you fully understand all the information that was given in Section Two. If there are any areas which you do not understand, re-read the relevant parts of the text.

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# 3 Operation of a DC Neutral Relay

## 3.1 Introduction

In this section you will look at the operation of the relay.

As was previously stated, a relay is an electrically-operated switch for opening and closing circuits.

*Forms of Relay*

In modern railway signalling, relays take a variety of forms according to the

- types of circuit in which they are to operate
- function to be controlled

## 3.2 Relay Operation

*Electro-Magnetic Field*

When the voltage is applied to the coil of the relay, **current flows** which produces an **electro magnetic field** in the soft iron core with

- a north pole at one end
- a south pole at the other end

*Normal Current Flow*

If the normal current route is adopted, the south pole passes through the yoke and then appears on the armature **opposite** to the pole face.

*Energisation*

Due to the fact that **opposite poles attract**, the armature moves on its pivot and is attracted to the north pole on the pole face of the soft iron core.

The relay is now said to be **energised**.

*De-Energisation*

When the voltage is removed from the coil, the **current stops flowing** and the electro-magnetic field **collapses**. There is no force to hold the armature against the pole face and it returns to its original position.

This action is sometimes aided by one or more **return springs**.

*Return Springs*

The return spring arrangement varies between the manufacturers. some depend entirely on the stiffness of the moving contact arms to achieve the return. Others add extra leaf springs pressing on the drive bar and some have an adjustable coil spring pressing on the armature.

When there is no current flowing the relay is said to be **de-energised**.

*Reversed Polarity*

If the voltage applied to the coil is reversed, the **current** flows in the opposite direction and the magnetic polarity is reversed, but the armature is **still attracted** to the pole face. This is a characteristic of the DC neutral relay, in which the relay will energise with the current flowing through the coil in **either** direction.

*Relay Coil Connections*

The R1 and R2 connectors on the plug board are each indicated by the symbol X on Figure 10. These are the **external** relay coil connections through which the relay is energised.

***Relay Coil Connections***

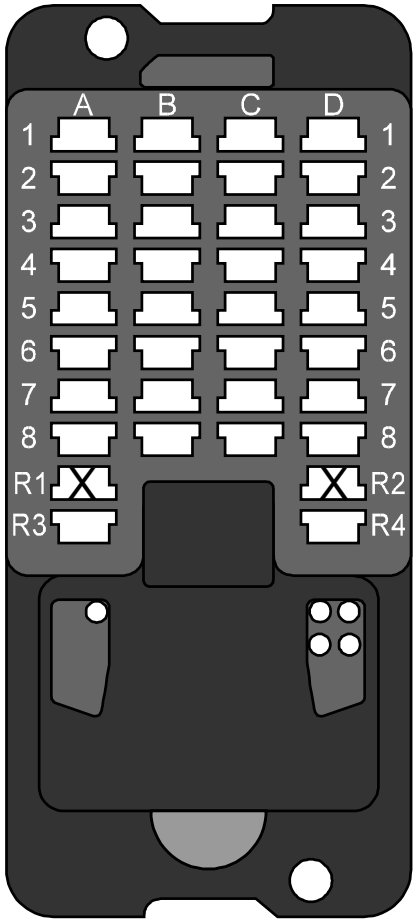



Figure 10 010

*Relay De-Energised*

Look closely at Figure 11 which provides a basic example of the relay coil and armature. You should recognise the component parts as these were examined in Section Two.

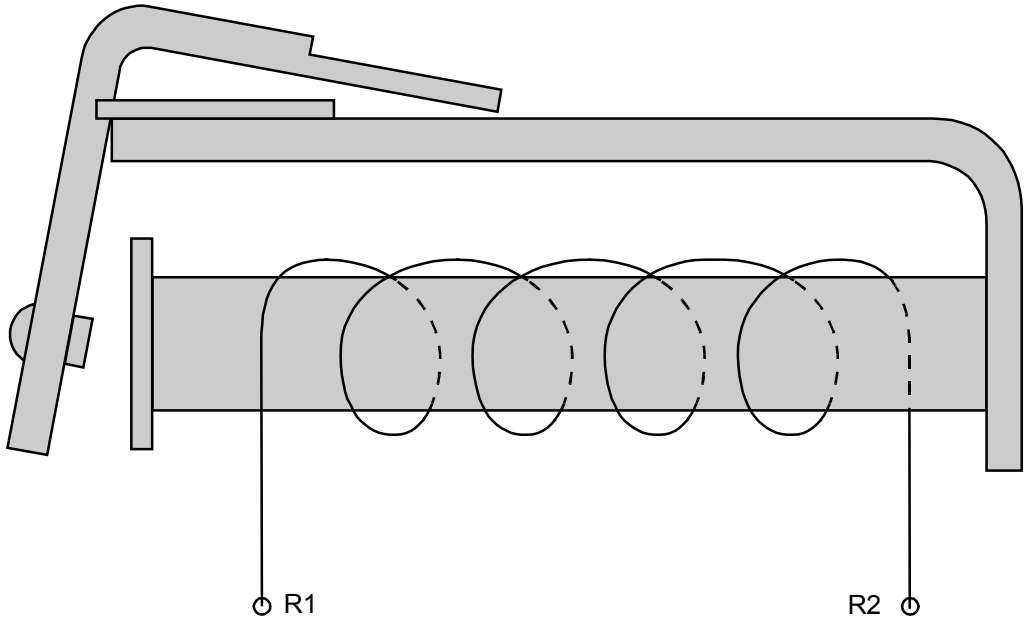



Figure 11

Figure 11 illustrates a relay when

- no voltage is applied
- there is no magnetic field

The relay is therefore **de-energised**

*Build Up of Magnetic Flux*

Now look at Figure 12 which shows the same relay with voltage present.

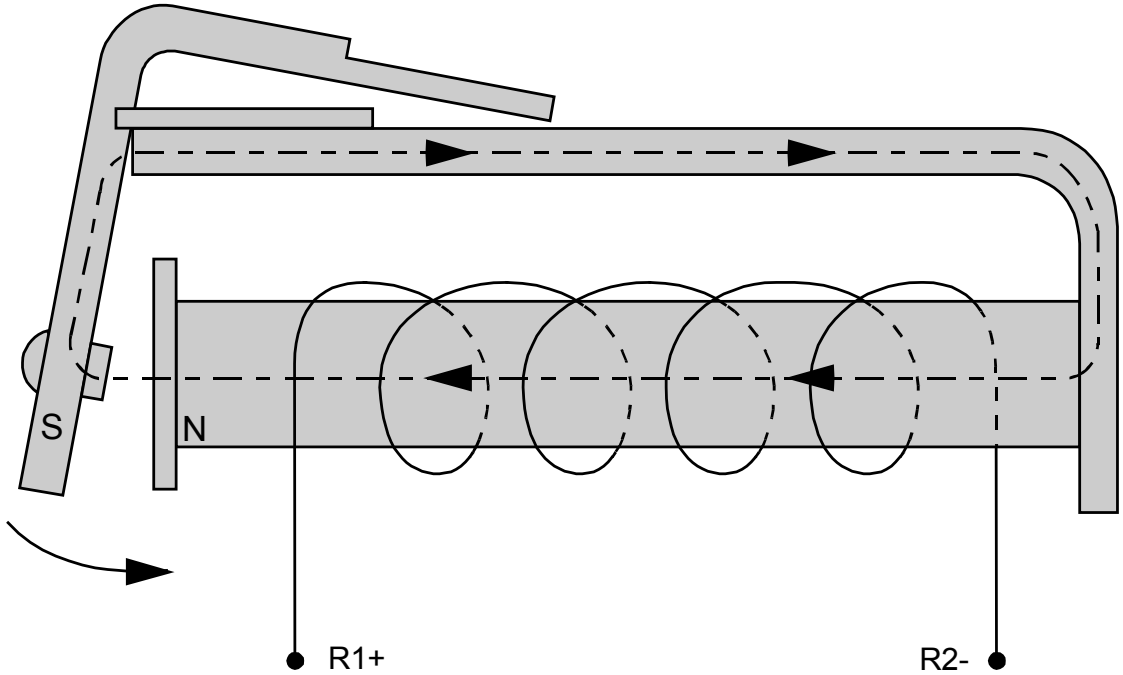



Figure 12

012

It can be seen that in Figure 12 the

- voltage is applied
- magnetic flux builds up
- armature starts to attract

*Relay Energised*

Now look at Figure 13 which illustrates the complete movement of the armature.

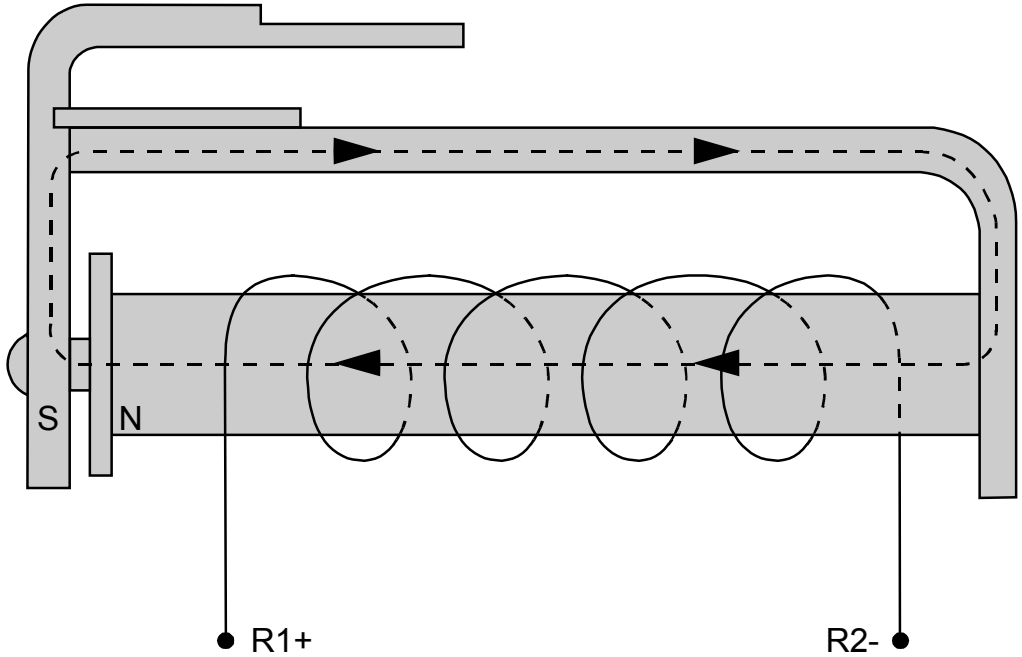



Figure 13

013

In Figure 13 it can be seen that the

- magnetic field is at full strength
- armature is attracted
- relay is energised

This process illustrates the basic operation of the relay.

It is now necessary to look at the part played by the **contacts** in this process. Before moving on to this, complete Self-Assessment Question 2.

## Self-Assessment Question 2

In your own words, state what is happening in Figure 14.

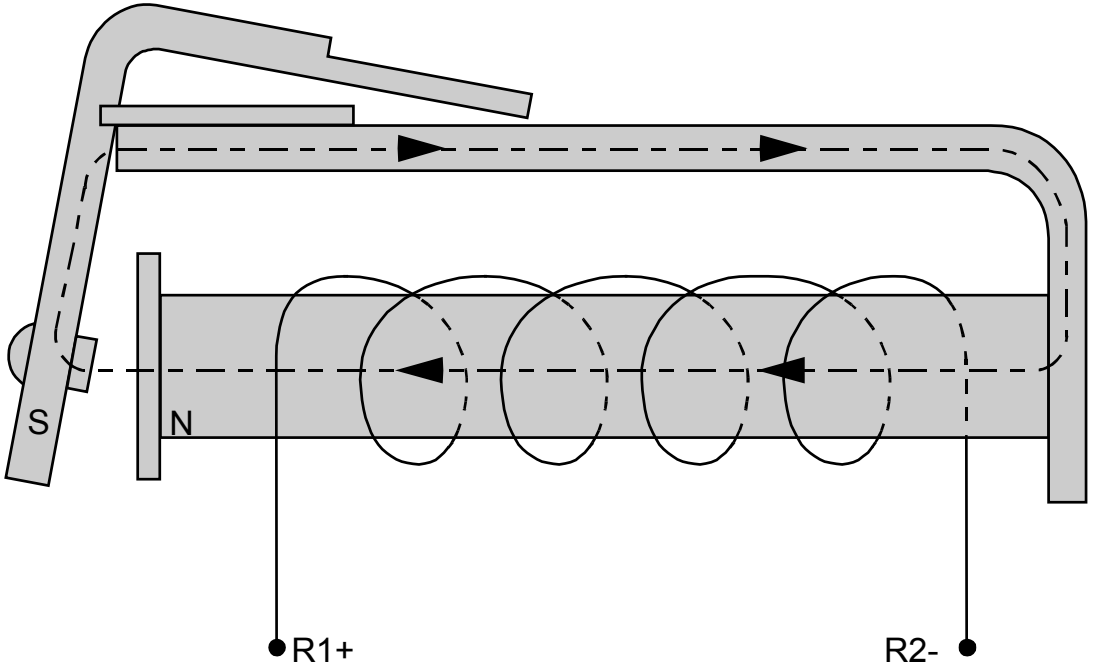



Figure 14

014

Now check your answer with the one given in the Answers Section. If you experienced any difficulty, re-read the text until you are confident you understand the information.

# Contact Operation

In Section One the contacts were described as a component part of the relay. You are now going to examine the actual operation of the contacts.

*Operation of the Contacts*

The description given in the Signalling Installation Handbook states that when the relay is energised

- back contacts break
- front contacts make

When the drive bar moves upwards the

- moving back contacts break with their associated fixed contacts
- moving front contacts make with their associated fixed contacts

*Contact Pair*

Each **contact pair** (one moving contact and one fixed contact) can be considered to be an **individual switch**.

Now look at Figure 15 which illustrates the position of the front and back contacts when the relay is **de-energised**.

*Relay Contacts in the De-Energised Position*

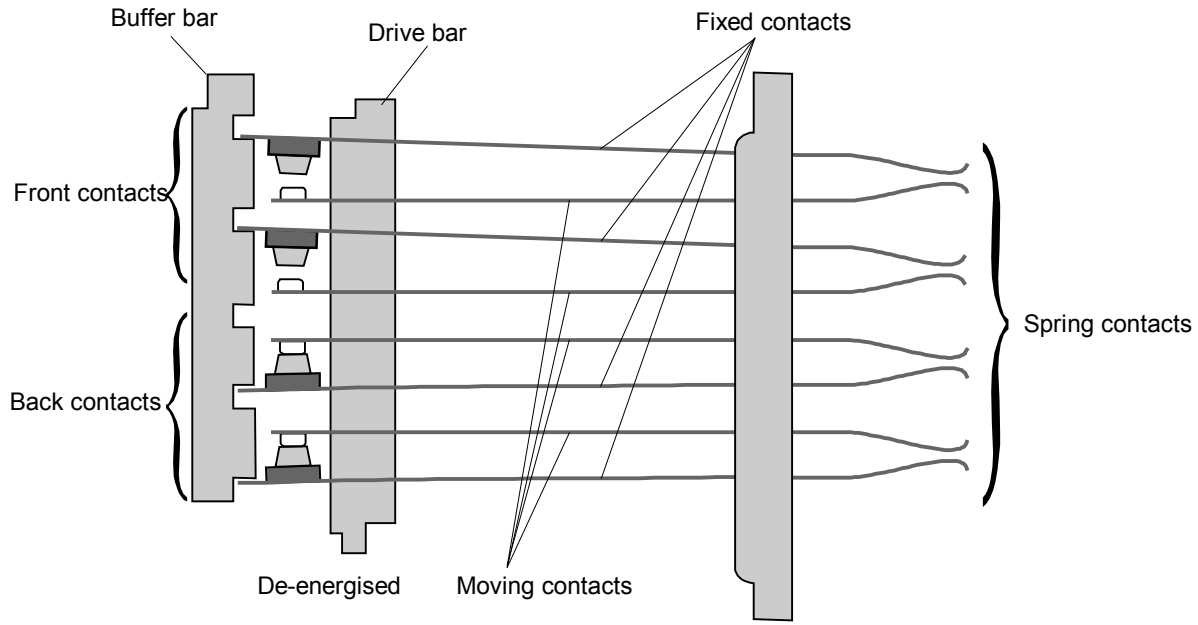



Figure 15

In Figure 15 it can be seen that the

- front contacts are broken
- back contacts are made
- relay is de-energised

Now look at Figure 16 which illustrates the position of the front and back contacts when the relay is **energised**.

*Relay Contacts in the Energised Position*

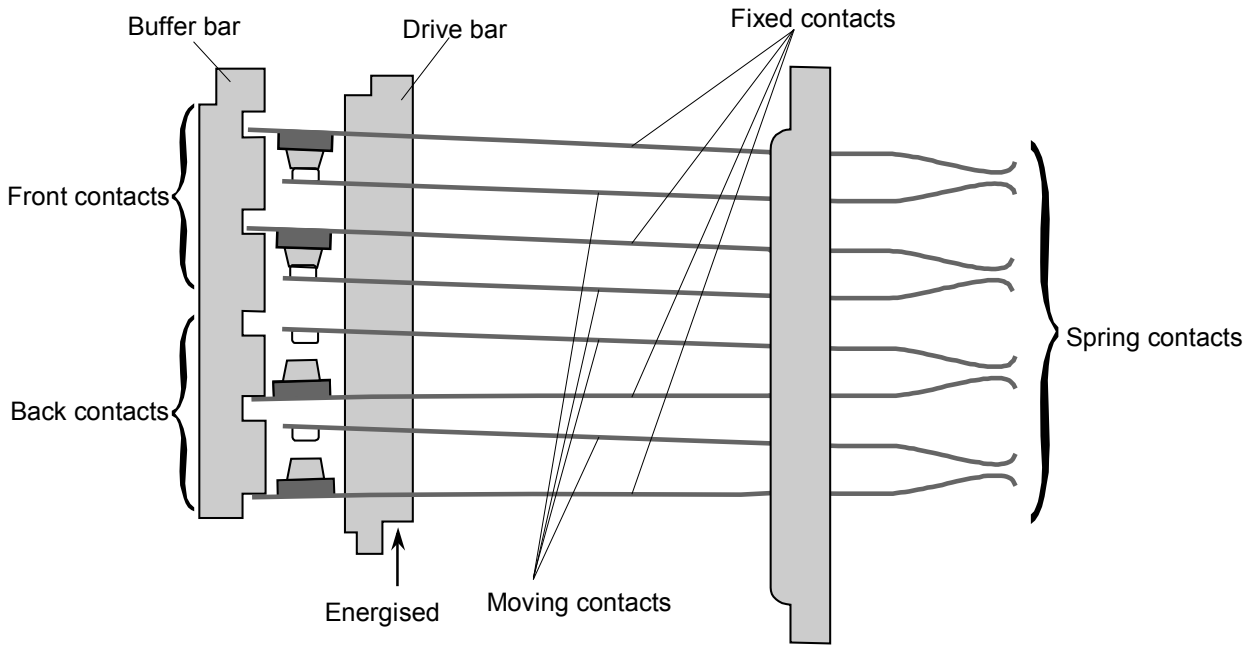



Figure 16

016

In Figure 16 it can be seen that the

- back contacts are broken
- front contacts are made
- relay is **energised**

It may be useful to look at the **glossary** at this stage to familiarise yourself with the alternative terminology which is used in connection with the contacts.

## 3.3 Section Summary

Section Three has looked at the operation of the relay.

It will be useful at this stage to refer to the Signalling Installation Handbook and read Section 2C 05.

Now complete Self-Assessment Question 3 before proceeding to Section Four.

## Self-Assessment Question 3

Figure 17 shows the contacts of row A of a relay and the equivalent written circuit symbols.

State which contacts are made when the relay is energised.

State which contacts are made when the relay is de-energised.

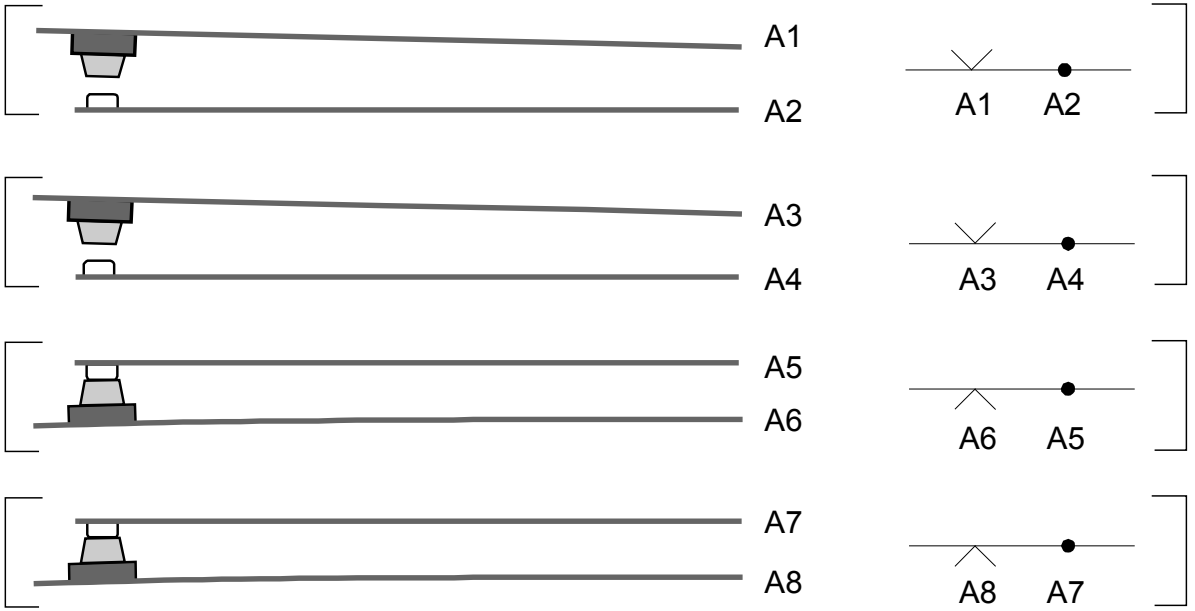



Figure 17

017

Now check your answer with the one given in the Answers Section. Before moving on to Section Four make sure that you fully understand all the information given in Section Three. If there are any areas which you do not understand, re-read the relevant parts of the text.

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# 4 The Main Relay Types

## 4.1 Introduction

In this section you will look at a number of main relay types in terms of

- physical characteristics
- operating differences

Look at Table 1 which illustrates some of the different types of relays in use.

| Type | Pin code | Contact arrangement | BR Spec | Typical uses |
|---|---|---|---|---|
| Neutral | 003 | 12F 4B | 930 | Line relay non-traction areas |
| AC Immune Neutral | 023 | 12F 4B | 931 | AC traction areas |
| Double Wound Neutral | 004 | 8F 8B | 930 | Controlling flashing aspects |
| Twin Neutral | 212 | 2 x 6F 2B | 960 | Line relay non-traction areas |
| Slow to pick (or slow to operate) | 041 | 8F 4B | 933 | Track repeating relay, route stick relays |
| Slow to release | 063 | 8F 4B | 934 | Point special relays, lamp proving repeat relays |
| Biased, Twin AC Immune | 049 | 2 X 6F 2B | 961 | Signal repeat relays, point control relays, point indication relays |
| Biased, AC Immune | 027 | 12F 4B | 932 | Point control relays, point indication relays |
| Magnetically latched | 011 | 11F 4B | 935 | Point lock relays, route lock relays |
| Polar stick | 014 | 8N 8R | 936 | Point lock relays |
| Contactor relay | 172 | 2HF 4B | 943 | Point contactors |
| **Timing Relays** | | | | |
| Synchronous motor | 189 | 1F 1B | 947 | Approach locking, track circuit time release |
| Thermal | 1053 | 2F 1B | 962 | Approach locking, track circuit time release |
| Electronic | 6047 | 1F 3B | 949 | Point operation |

Table 1

Most relays have 50 V DC coils but a few associated with special applications operate on lower voltages.

As you can see from Table 1 each relay has a

- unique pin code number
- specific arrangement of the contacts
- BR specification number

You will now look at the relays listed in Table 1 in greater detail.

Note some relays may have different contact arrangements with the same pin code.

## 4.2 Neutral Relay

The DC neutral relay has already been examined in Section Three.

## 4.3 AC Immune Relay

With the introduction of AC railway electrification it was found that faults could arise in DC circuits, which consisted of unwanted AC appearing by induction in the DC supply to the relay. This could cause

- false operation
- wrong-side failures

This means that in AC areas, the relays need to be immunised against operation by AC.

The AC immune relay has a copper sleeve inserted over the core adjacent to its pole face. The copper sleeve is usually referred to as a **copper slug**.

When the DC voltage is initially applied to the coil

- the slug delays the build up of the magnetic flux
- the response time of the relay is increased

If the normal mains frequency AC is applied to the coil, the build up and drop away of flux due to the AC current is much faster than the response time of the relay. The result of this is that the armature does not respond.

Figure 18 provides an example of an AC immune relay.

*AC Immune Relay*

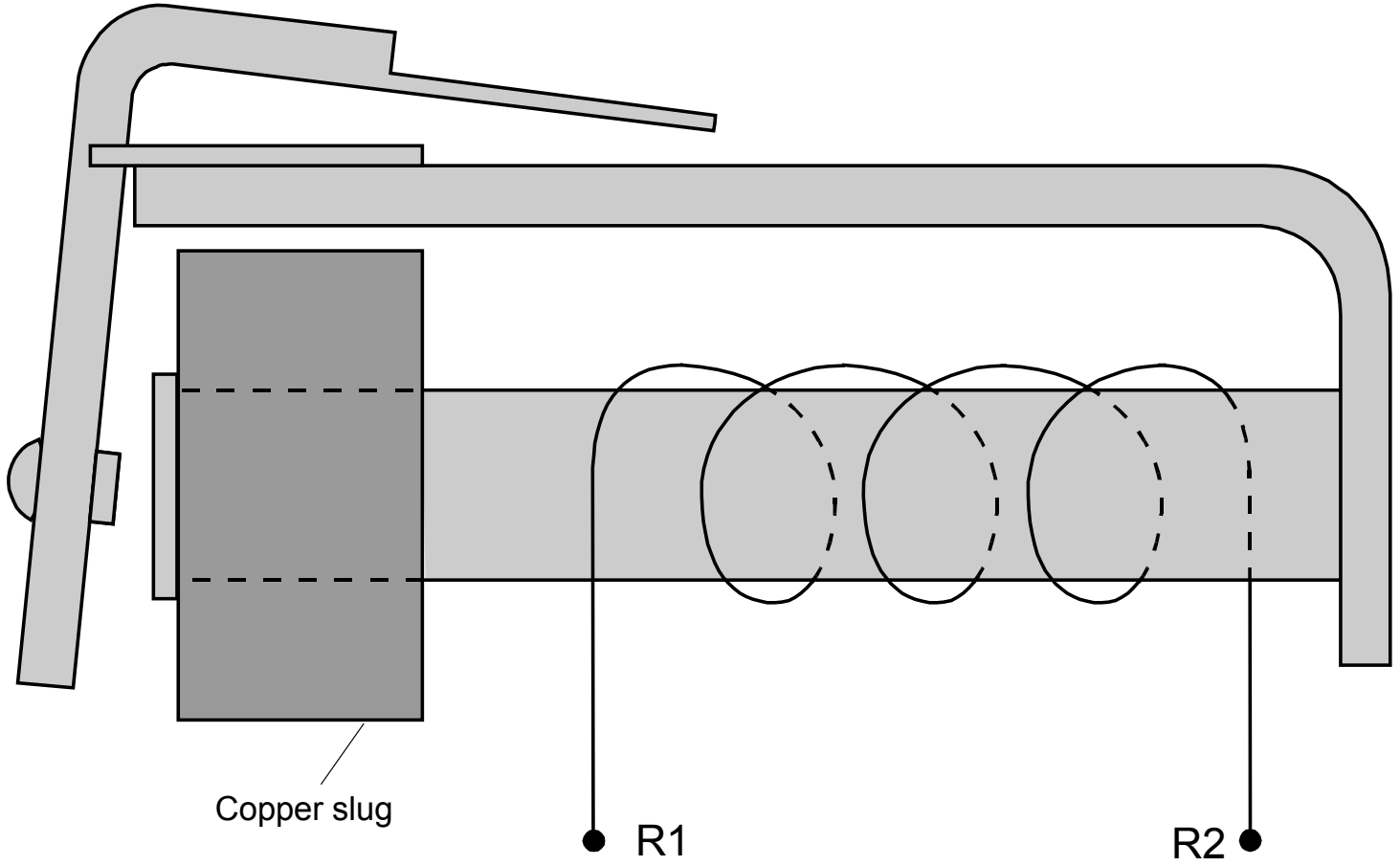



Figure 18

019

## 4.4 Double Wound Relay

The BR 930 style relay can have two identical but electrically separate windings on the same core. If current is applied to either coil it will energise the relay. This is a double-wound relay with

- R1/R2 for one coil
- R3/R4 for the other coil

Due to the fact that this relay has two windings it can be controlled from two separate circuits.
Figure 19 shows a double wound relay.

*Double Wound Relay*

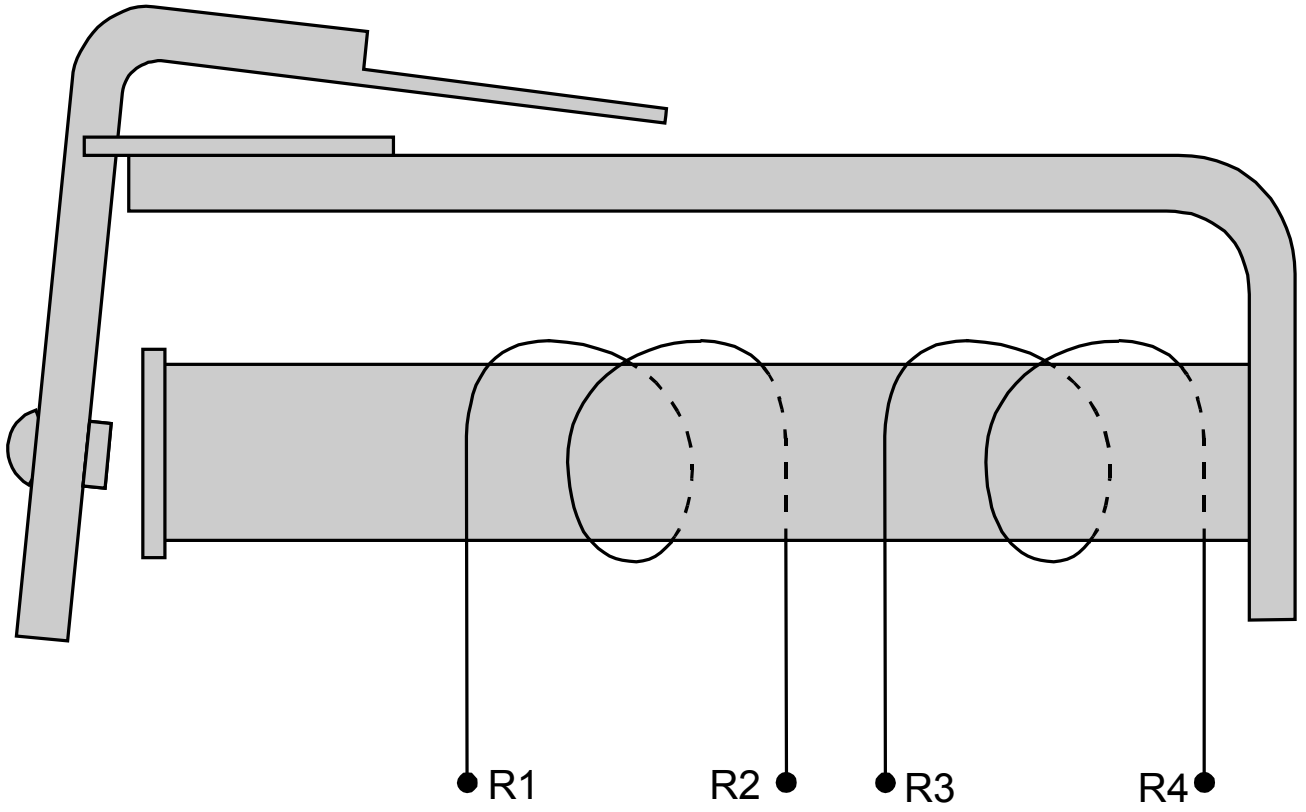



Figure 19

020

## 4.5 Twin Relay

The twin relay accommodates the equivalent of two single relays, carrying half the usual number of contacts within the width of a single relay case.

The basic advantages of a twin relay are as follows

- reduced cost when compared to the purchase of two separate relays
- reduction of space required in a relay room or location case
- reduced installation costs

Figure 20 illustrates the arrangement of a specification 961 AC immune twin biased relay.

Biased relays will be covered later in this section.

*Example of a Twin Relay (Spec 961)*

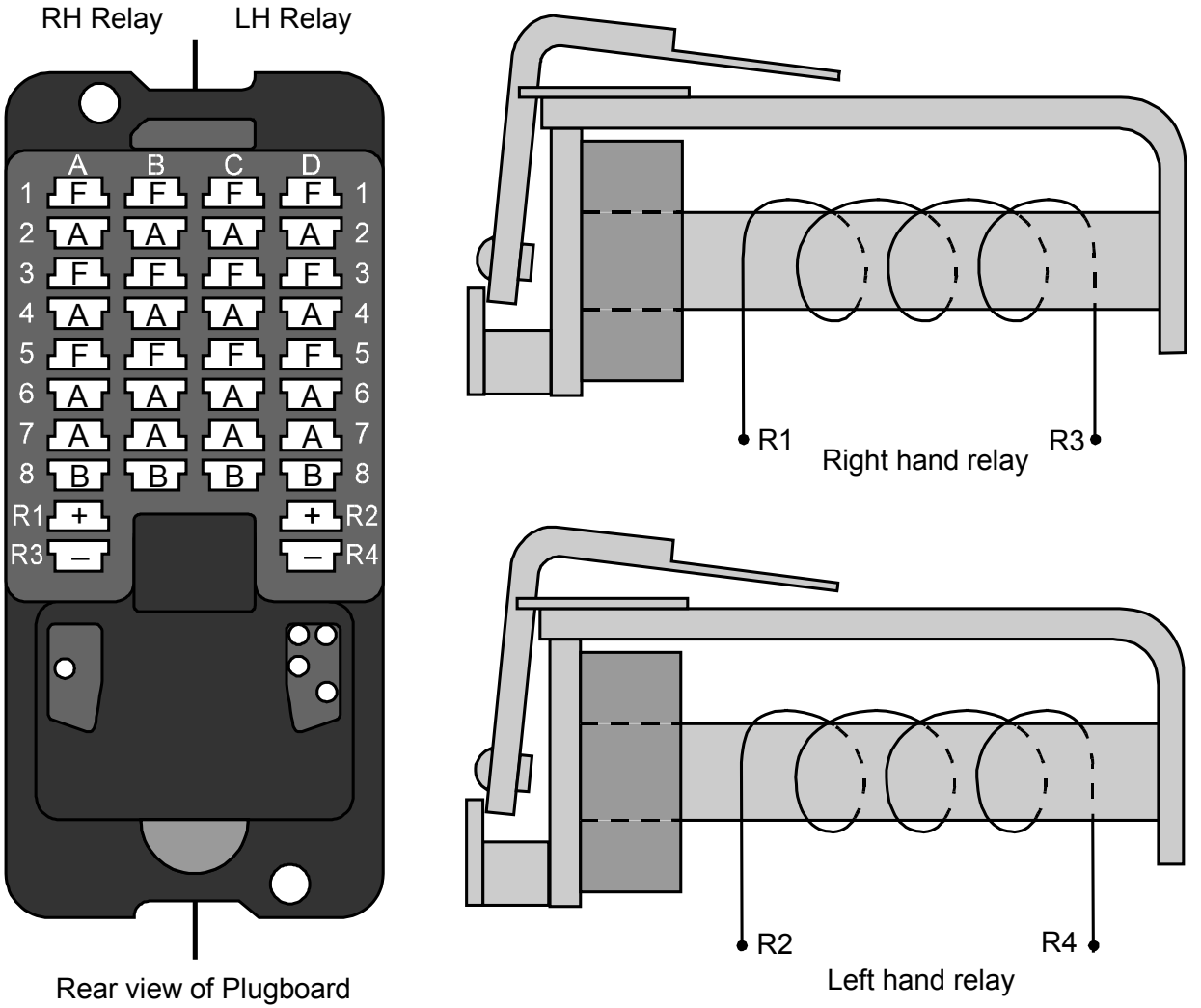



Figure 20

## 4.6 Slow to Pick Relay

In the slow to pick relay, a large copper slug is used which is placed at the **yoke** end of the core.

In addition a **magnetic shunt** is introduced to avoid the relay taking on the characteristics of a slow to release relay. The delayed pick up time is 400 ms (milliseconds).

The slow to pick relay is also **AC immune** due to the effect of the copper slug.

An example of the slow to pick relay is shown in Figure 21.

*Slow to Pick Relay*

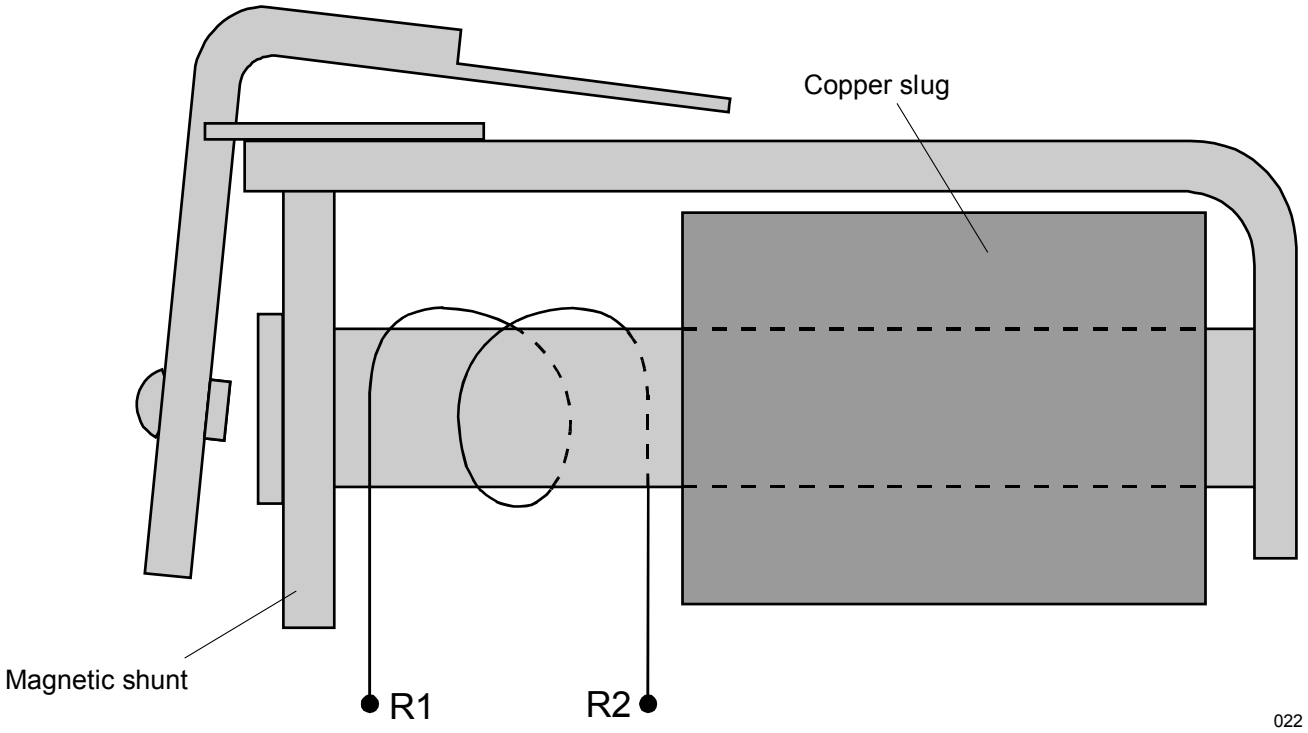



Figure 21

## 4.7 Slow to Release Relay

In the slow to release relay, increasing the size of the **copper slug** normally found in AC immune relays achieves the **slow to release characteristic**. The delayed release time is 200 ms.

It must be remembered that the copper slug also

- produces the slow to pick characteristic
- makes the relay AC immune

An example of the slow to release relay is shown in Figure 22.

*Slow to Release Relay*

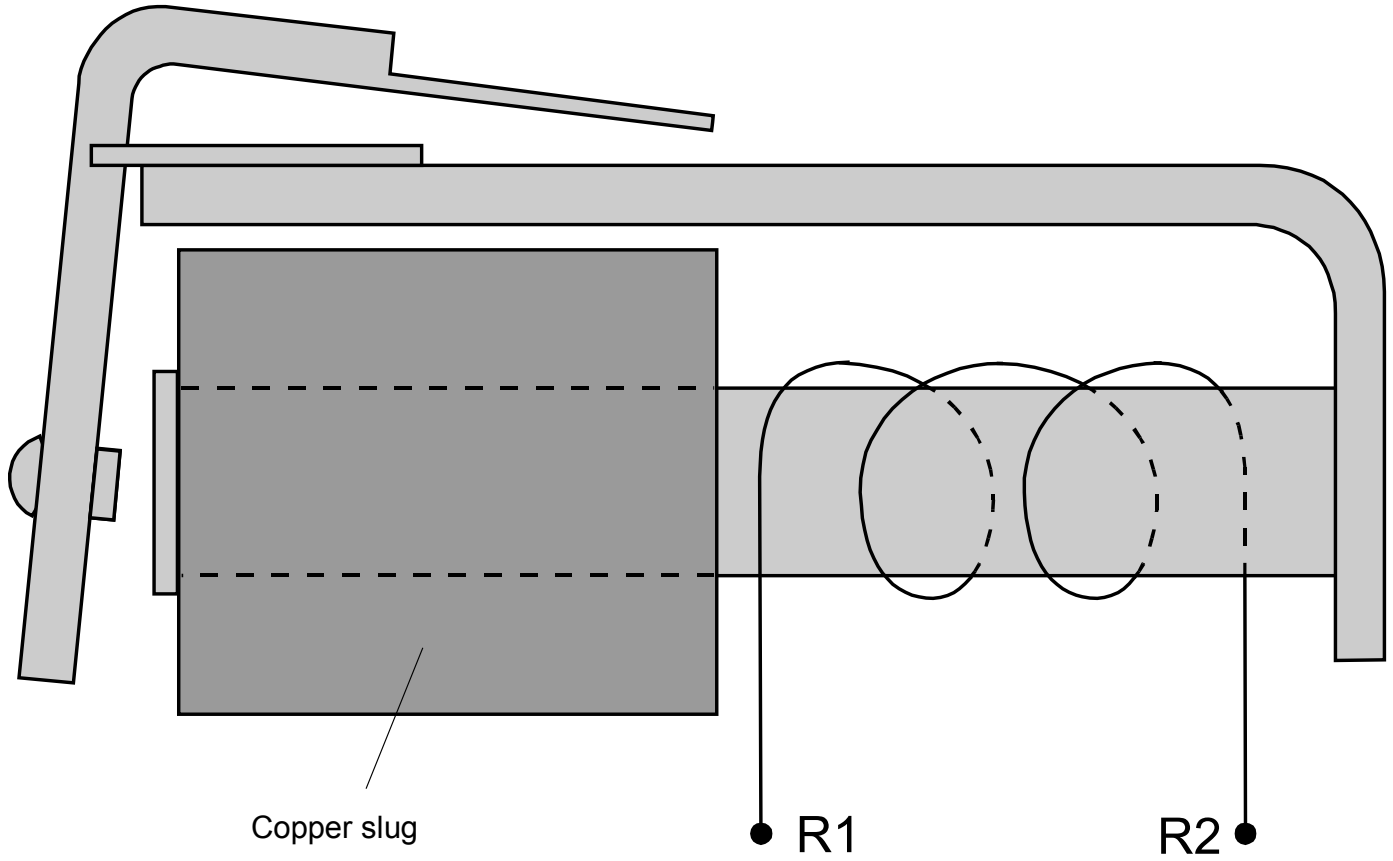



Figure 22

023

## 4.8 Biased Relay

A biased relay should only pick with the current in **one direction** through its coil.

In order to ensure that the biased relay incorporates this characteristic it utilises one of the following methods

- a series magnet attached to the pole face
- a shunt magnet used to bridge the core and heel piece
- a series magnet attached to the armature

If this type of relay needs to be AC immune a copper slug is also introduced.

An example of a biased relay using a series magnet attached to the pole face is shown in Figure 23.

The relay will

- energise if R1 is positive and R2 is negative
- not energise if R1 is negative and R2 is positive

This feature is included in a relay so that, for example, the normal and reverse control relays for a set of points at a remote location can be operated over a single pair of conductors.

*Biased Relay*

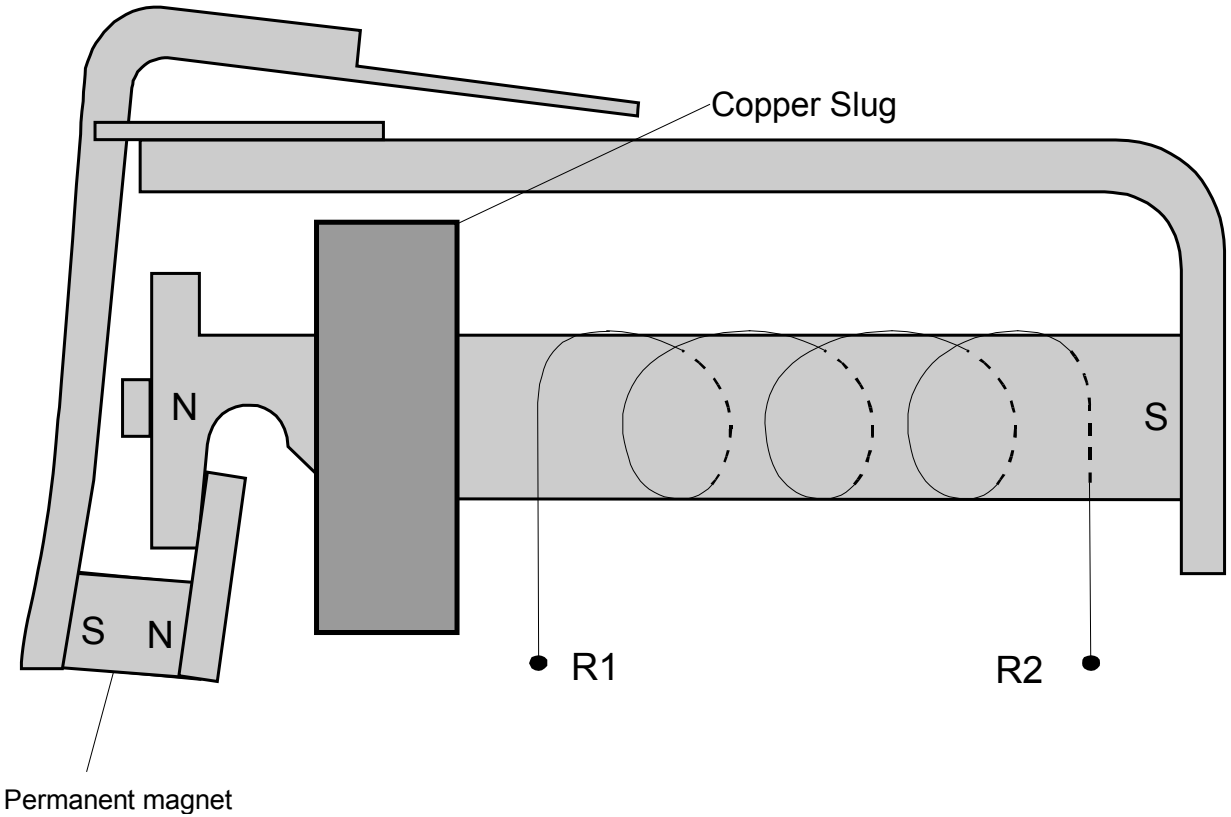



Figure 24

## 4.9 Magnetically Latched Relay

In the magnetically latched relay, if the power is removed the armature remains in its last position until further power is applied to change its position.

It has two coils

- one wound conventionally to pick up (latch) the relay
- the other wound in the opposite direction to release (un-latch) the relay

This type of relay does not require a residual pin. It is used in inter-locking circuits where it is important that the contacts do not change state in the event of a power supply failure.

Figure 24 provides an example of a magnetically latched relay.

It can be seen in figure 24 that the relay will

- latch if R1 is positive and R2 is negative
- un-latch if R3 is positive and R4 is negative

One of the relay front contacts is connected (internally) in series with the release coil. This is to ensure that once the armature starts to drop away the electrical circuit is disconnected, ensuring that the magnetic flux cannot build up sufficiently to re-pick the relay. This contact is known as the cut off contact.

*Magnetically Latched Relay*

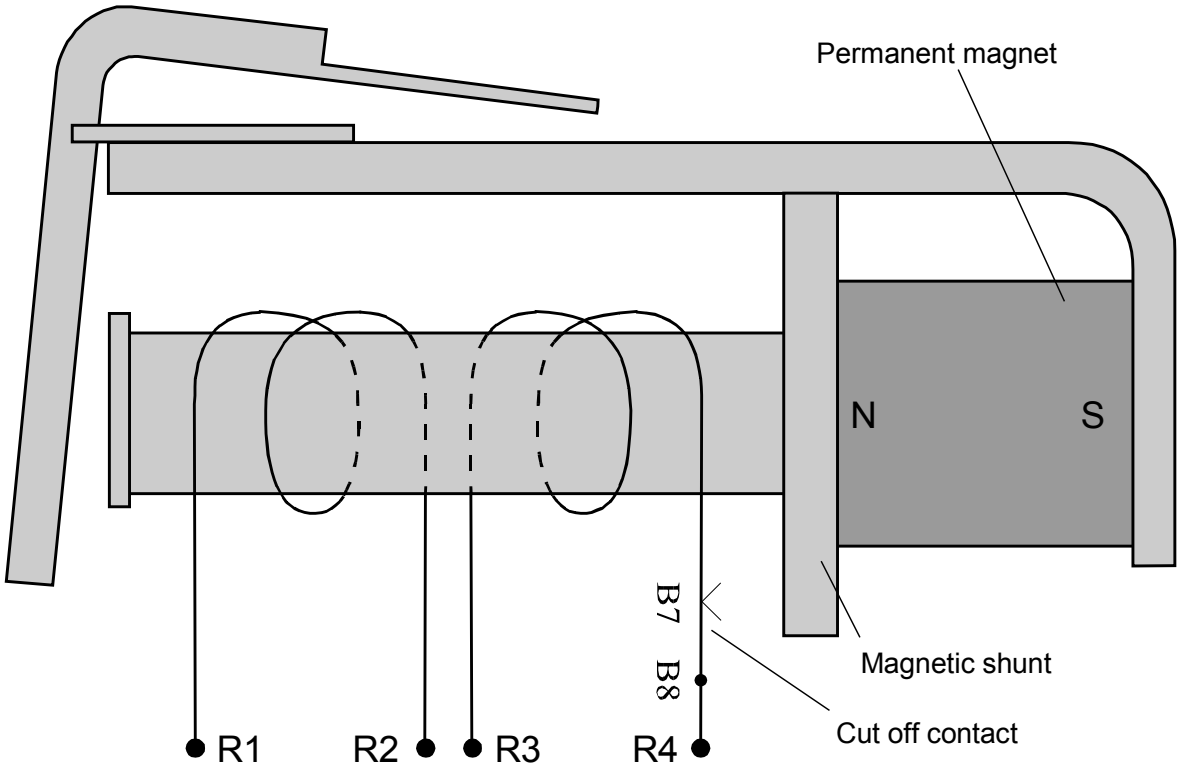



Figure 24

## 4.10 Polar Stick Relay

Figure 25 provides an example of a polar stick relay.

It can be seen in Figure 25 that the construction of the polar stick relay is similar to the biased relay except that

- there is no residual stud
- the soft iron core is shaped to make contact with the moving soft iron armature extension in both pick up and release positions

This has the effect of making the relay **stick** in the last position called. Feed applied to the coil makes the armature move in one **direction** and it is held in position **magnetically** if the power is removed.

Reversal of the coil feed results in the armature moving to the opposite side, where it is again held.
Like the latched relay it contains two coils and can give various permutations of operation

- either coil can control the relay individually using the current reversal as described above. Alternatively both coils controlled by separate circuits can be achieved
- using one coil to pick and the other to release just like the latched relay. This is the most common arrangement.

*Polar Stick Relay*

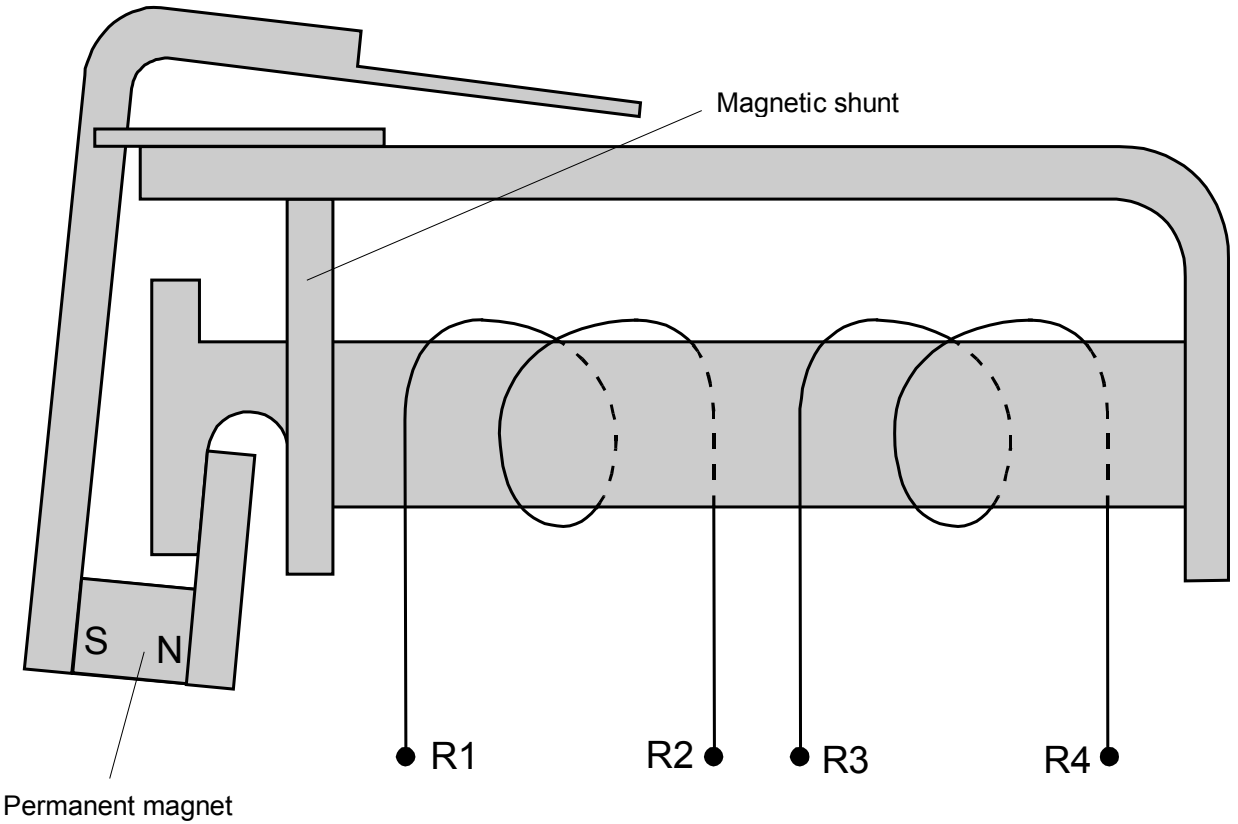



Figure 25

026

# 4.11 Contactor Relays

For operating point machines and other heavy current loads, the standard contacts on a series 930 relay are inadequate and a special contactor relay is employed. In practice, the contactor relay is of biased AC immune construction and has four standard back contacts, but has two heavy duty, silver carbon front contacts, each fitted with a magnetic **blow-out**.

The blow-out arrangement consists of a magnet located so that the spark generated when the contacts break is **blown** outwards and quickly extinguished before burning of the contacts can occur. To improve the breaking capacity the pair of front contacts are connected in series, as illustrated in Figure 26. This is known as **double cutting**.

*Utilisation of Contactor Relay*

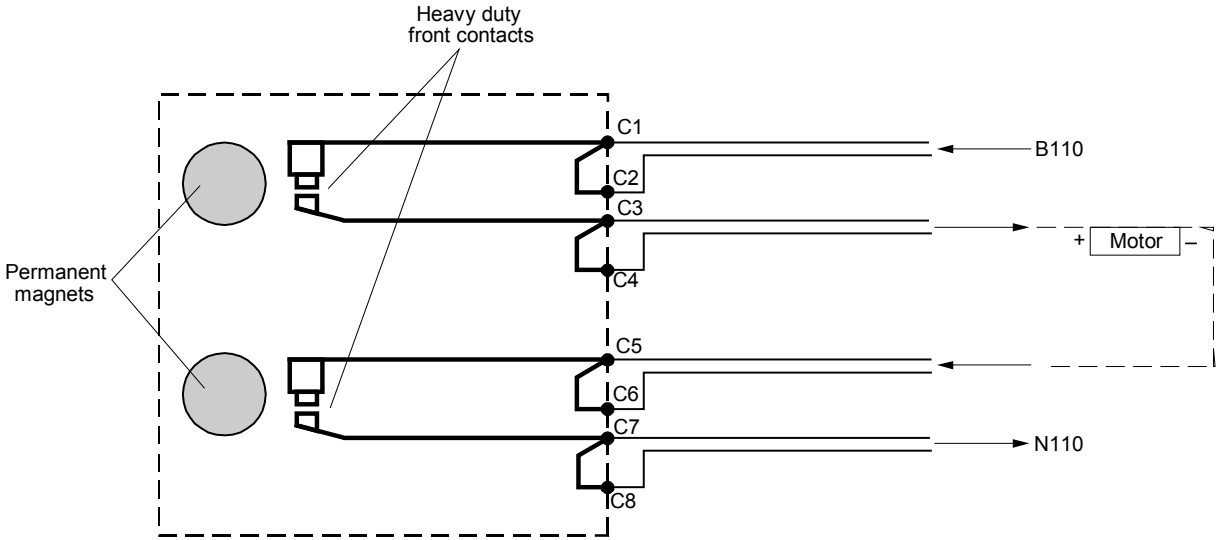



Figure 26

## 4.12 Timing Relays

There are a number of relays which can be placed under the general heading timing relays, including

- synchronous
- thermal
- electronic

Each of these will be examined separately.

### Synchronous Motor Relay

This unit consists of a synchronous AC motor, which operates a pair of contacts via a gear train.

The motor is normally 110 V AC fed. The delay time may be altered by changing the distance the gears have to move before they operate the delayed contacts. When the delay is completed the motor is de-energised and the gear train is driven by a coil spring back to its start position. Depending on the time delay required, the following timer relays can be used

- 30 - 240 seconds, code 189
- 2 - 50 seconds, code 191

Meaningful illustration of the physical construction of a synchronous motor relay is very difficult to show. It is easier if you ask your supervisor or your mentor to show you an actual example.

*Synchronous Motor Timer Relay Circuit*

Figure 27 illustrates the internal circuit of a typical synchronous motor relay.

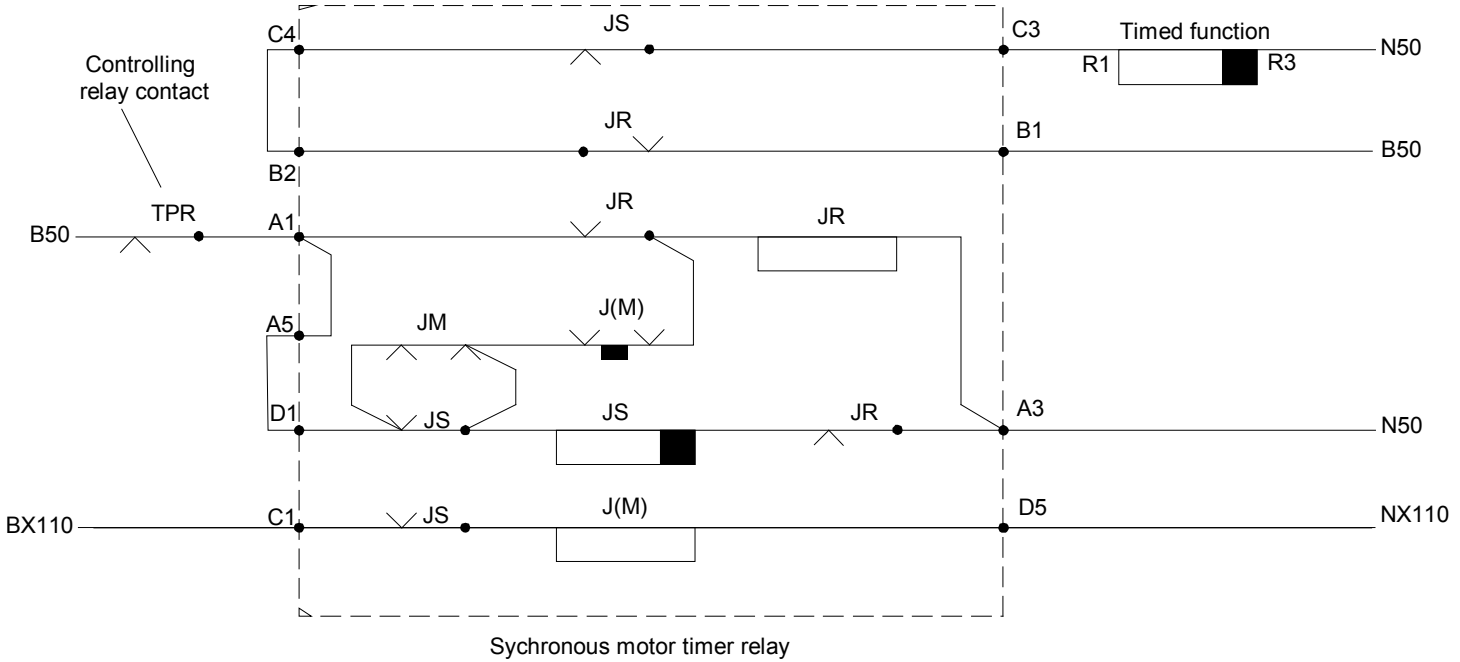



Figure 27

028

An explanation of how the timer relay circuit operates is as follows

- the synchronous motor J(M) has a back contact (marked J(M)) which is closed when the motor is de-energised and the coil spring has driven the gear train back to its start position. It also has a front contact (marked J(M)) which is closed when the motor is energised and has driven the gear train for the duration of the delay period
- incorporated in the relay assembly are two dc relays, JR is a neutral type and JS is a slow to release type
- the starting condition is when the TPR contact is open, JR and JS are de-energised and the timed function relay is de-energised
- when TPR contact is closed, relay JS is energised via contact JM and the back contact of JR. JS then sticks via its own front contact and energises the motor J(M). Contact JM opens and the pre-set delay period commences
- when the delay expires, contact J(M) closed and energises relay JR, which then sticks via its own front contact
- when relay JR is energised, its back contact de-energises relay JS
- when relay JS drops it de-energises the motor J(M), which allows the gear train to return to its start position, and the timed function relay is energised via the front contact of JR and the back contact of JS
- relay JR and the timed function relay then remain energised until the TPR contact is opened

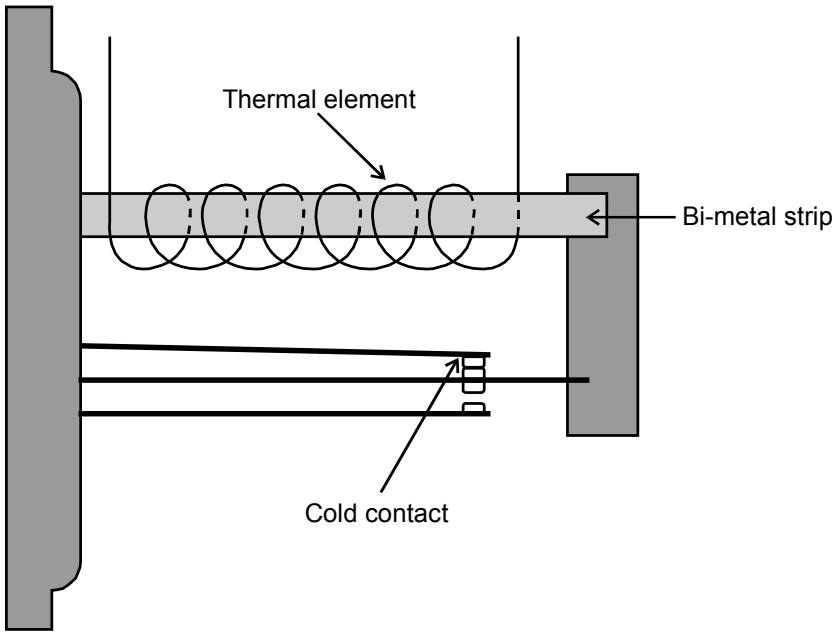
## Thermal Timer Relay

The thermal relay uses a bi-metallic strip which bends with heat and then cools down. Each time this occurs it closes a contact. The timing is complete once the **cold contacts** have made.

Illustrations of this type of relay are shown in Figures 28, 29 and 30

Figure 28 shows the element cold and the **cold contact made**.

*Thermal Element Cold*

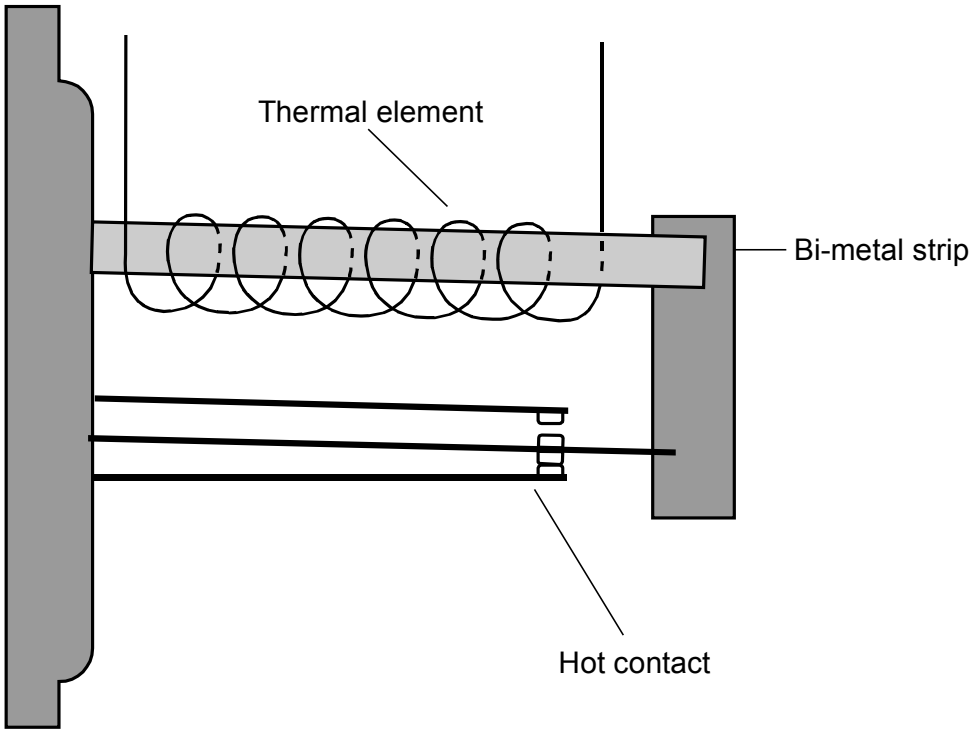



029

Figure 28

*Thermal Element Hot*

Figure 29 shows the element hot and the **hot contact made**.




030

Figure 29

*Circuit for a Thermal Timer Relay*

Figure 30 shows internal circuitry for a thermal relay.

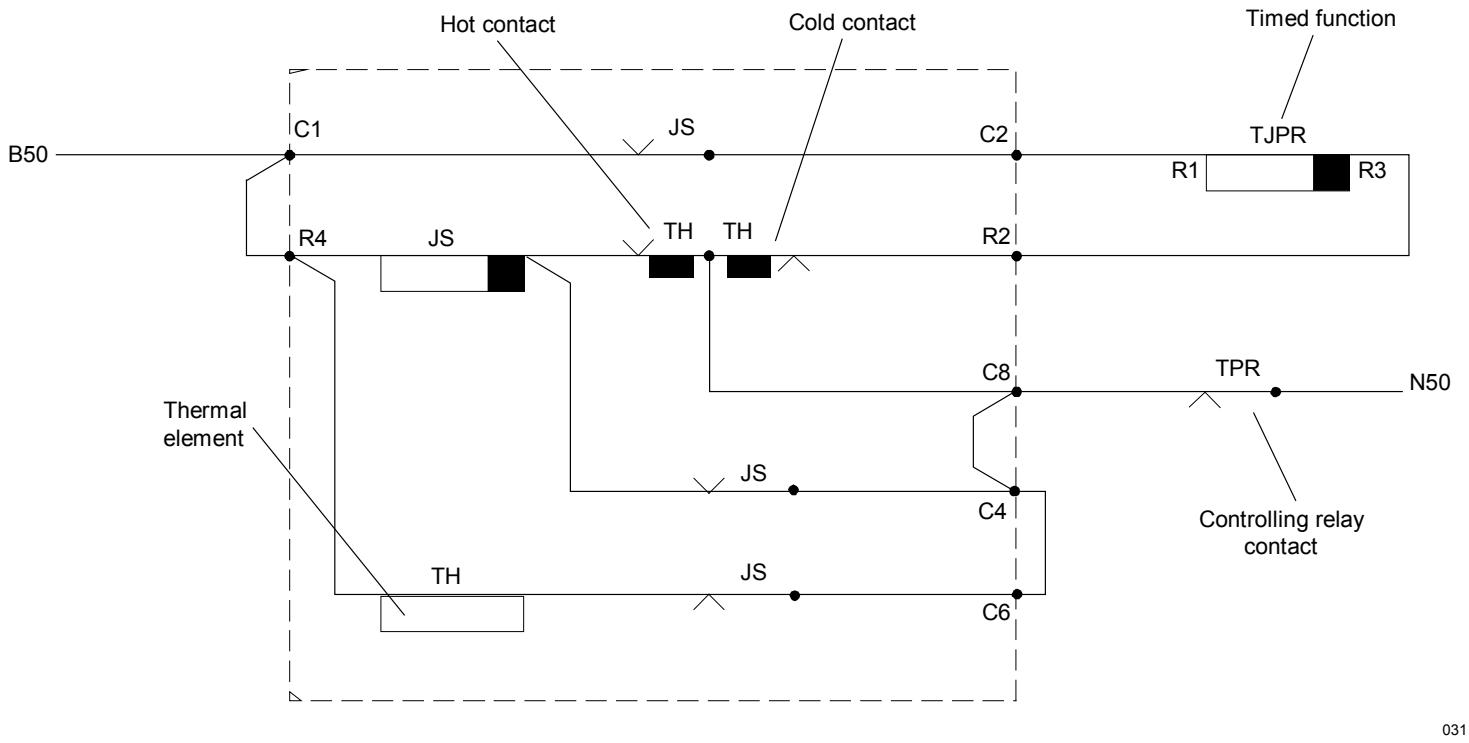



Figure 30

An explanation of how the thermal timer circuit operates is as follows

- initially the controlling relay contact TPR is open, the thermal element is cold, the slow to release relay JS is de-energised and the timed junction relay TJPR is de-energised
- when contact TPR is closed the thermal element is energised
- as the bi-metallic strip warms up the cold contact opens and after a further delay the hot contact closes and energises relay JS
- when relay JS picks it sticks via its own front contact and its back contact breaks, de-energising the thermal element
- when the bi-metallic strip has cooled down, the pre-set delay expires when the cold contact closes and relay TJPR is energised via the cold contact and a front contact of relay JS
- relays TJPR and JS then remain energised until the contact TPR is opened.

There are additional relay types which are used in specific circumstances. Information concerning these relays can be found in The Signalling Installation Handbook.

## Electronic Timer Relay

This form of time delay unit is usually used as the points timer in point control circuitry. It is used to limit the time the point machine motor runs to about 7.5 seconds, which is achieved by using a capacitor/resistor network. This delays the build-up of flux in a Post Office 3000 relay. The basic circuit arrangement is shown in figure 31.

*Basic Circuit for Electronic Timer Relay*

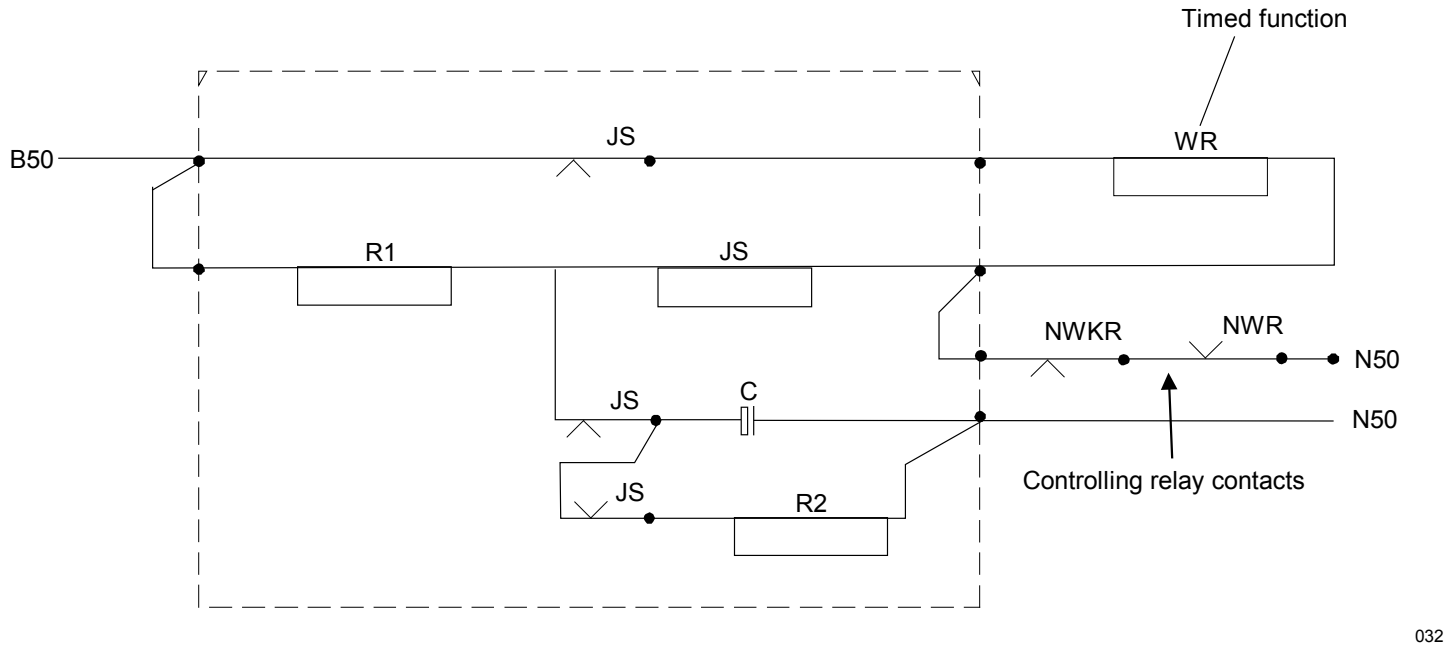



Figure 31

An explanation of how the electronic timer circuit operates is as follows

- initially the controlling relays NWKR and NWR are both de-energised and
the JS relay is de-energised
the WR relay is de-energised
the capacitor C is discharged

- when the points are called to normal, the NWR picks and
WR picks via a back contact of JS
the capacitor C starts to charge up via R1 and another back contact of JS

- as C charges up, the voltage across the coil of JS builds and
after about 7.5 seconds JS picks and is held up via R1
C begins to discharge via R2
WR is de-energised

- JS remains energised until NWKR is energised (indicating the points are detected normal) or the points are called to reverse and NWR drops.

In practice, it is normal for the points to complete their travel and NWKR to be energised before the 7.5 seconds has expired. This immediately de-energises JS and WR.

When the points are called reverse the controlling relay contacts are RWKR and RWR.

# 4.13 AC Vane Relays

AC Vane Relays are used with AC track circuits to detect the presence of AC Currents. The DC track circuits use special versions of the series 930 or other standard types of neutral or AC immune relays to detect the DC currents. Detection of AC currents in track circuits requires a totally different method, which is sensitive to the magnitude, the frequency and the phase of the alternating current. In addition, the AC detector should reject

- DC
- unwanted AC of a different frequency

## Construction of an AC Vane Relay

The basic construction of a double element vane relay is shown in figure 32. The vane is pivoted to move in the gaps of the cores of two electro-magnets. One electro-magnetic core is referred to as the local core and the other as the control core.

The torque (force) acting on the vane at any instant is provided by the magnetic flux from the local core acting on the eddy currents produced in the vane by the control core **and** the magnetic flux from the control core acting on the eddy currents produced in the vane by the local core.

When the two cores are energised by AC of **different** frequencies the torque is **alternating** at a frequency equal to the difference of the two frequencies and produces no net movement of the vane.

Only when the currents in both electro-magnetic coils are of exactly the same frequency is there any steady torque applied to the vane. Then the torque is proportional to the magnitude of the two AC currents and is a maximum when their phases are in quadrature (that is the phase difference is 90°). When the phase difference is zero or 180°, the torque is zero. Of course, the torque is zero when the AC current in either coil is zero.

*Double Element Vane Relay*

The purpose of the slots in the vane is to concentrate the eddy currents and thereby increase the torque for a given level of excitation current in either electro-magnetic coil. To produce the relay action, the movement of the vane is used to move conventional relay contacts. It is normal for the local core to be energised by 110 volts at the required AC frequency and for the AC operating current to be applied to the control core.

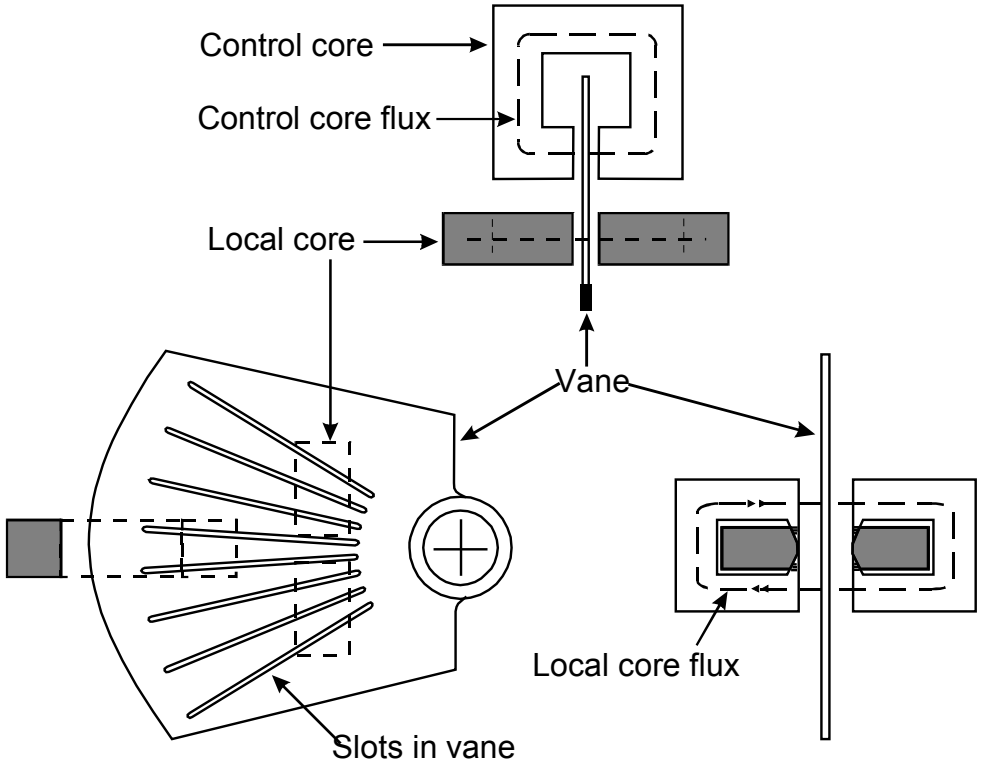



Figure 32 03

## 4.14 Section Summary

This section has looked at the following types of relay

- neutral
- AC immune
- double wound
- twin
- slow to pick
- slow to release
- biased
- magnetically latched
- polar stick
- contactor
- timing relays
  - synchronous motor
  - thermal
  - electronic
- AC vane relays

Now complete Self-Assessment Question 4 before proceeding to Section Five

## Self-Assessment Question 4

Correctly identify the relay shown in Figure 33. Explain where it would be appropriate to use this relay.

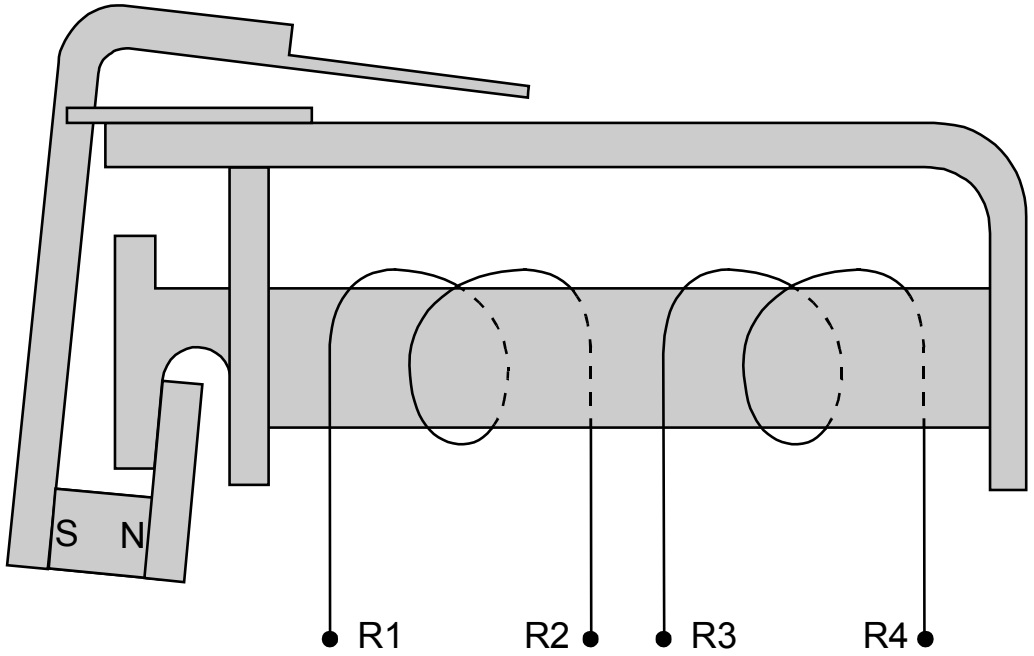



Figure 33

032

Now check your answer with the one given in the Answers Section. Before moving on to Section Five make sure that you fully understand all the information given in Section Four. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 5 The Relay Room and Location Cases

## 5.1 Introduction

Up to this stage we have looked at

- the components of a relay
- relay types

You need to be able to locate specific relays within the relay room. You must also be able to identify the relay type and know how to locate the relay contacts.

Section Four examined the main types of relays. In this section you will look at where the relay components are located within the relay room.

## 5.2 Relay Room Layout

The layout of the relay room will be in accordance with the diagrams issued by the Design Office. These diagrams contain the

- relay room plan
- individual profiles

The relay room plan provides the relative position of equipment racks within the structure.
A typical plan of a relay room will show both

- orientation
- equipment location

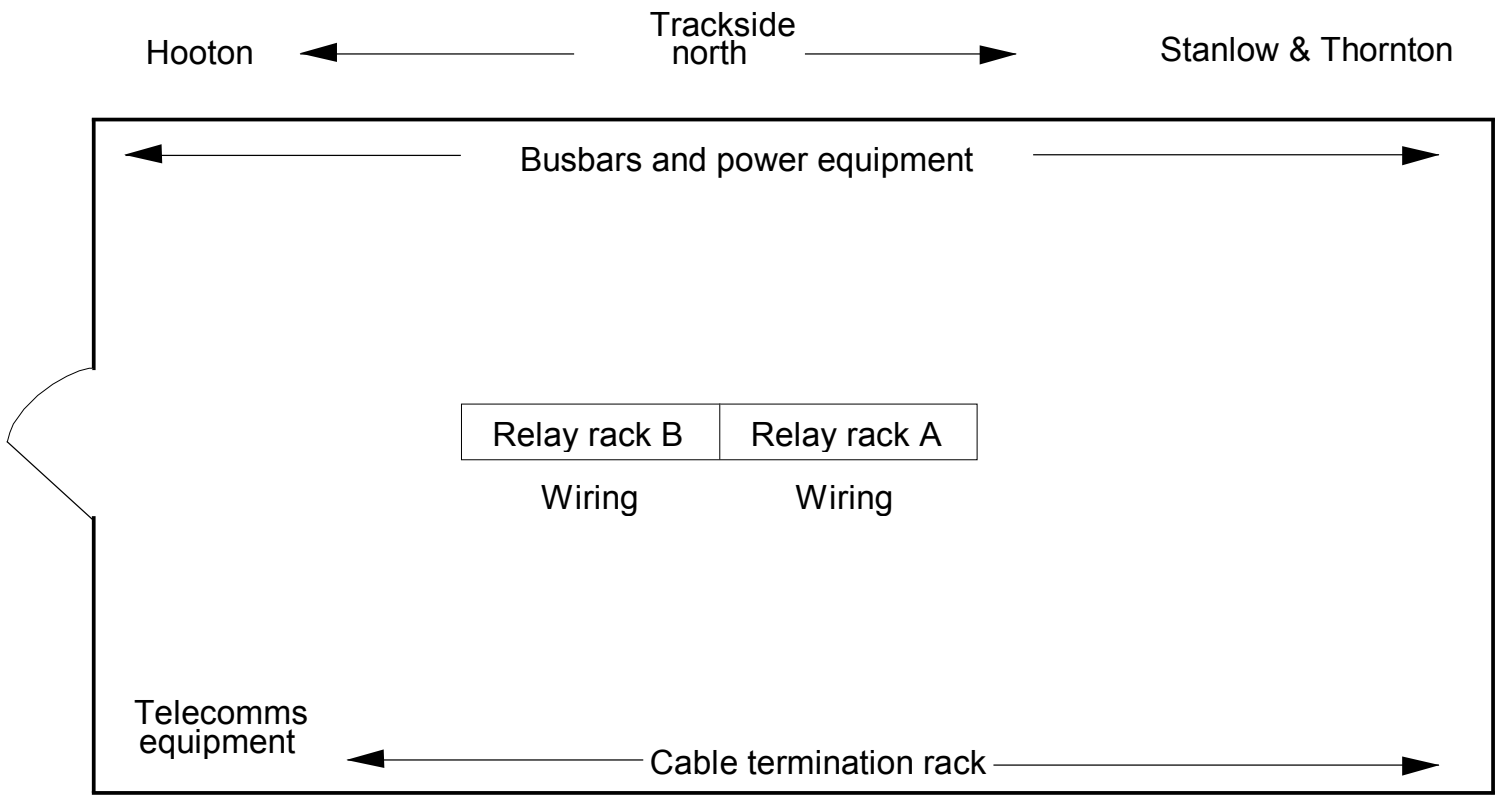
As can be seen from Figure 34 the relay room contains the following

- relay racks
- power equipment racks
- cable termination racks

and in some cases telecomms equipment.

Here you will only deal with the relay racks and the directly related equipment.

*Typical Relay Room*




033

Figure 34

## 5.3 Relay Rack Profile

The **relay rack profile** details the equipment on each individual rack, covering the relays and the associated equipment.

The relay rack profile gives the **position** of all relays and equipment.
The profile is viewed from the front

- the rows number from top to bottom
- the columns number from left to right

Test points are usually provided on each relay rack, in which case the allocated position is also shown on the rack profile.

The following information is also provided on the rack profile for each relay

- relay reference
- BR Spec number
- pin code
- contact arrangement

A typical example of a relay rack profile is shown in Figure 35.

*Typical Relay Rack Profile*

RackA

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Up main<br>BCR | Down main<br>`LCR' BCR | Down main<br>`TOL' BCR | | | | | T1<br>TxR | T1<br>Tx0 | T19<br>TxR | | T19<br>Tx0 | | |
| T1<br>T2PR<br>933<br>043<br>8F4B | T2<br>T2PR<br>933<br>043<br>8F4B | T3<br>T2PR<br>933<br>043<br>8F4B | T4<br>T2PR<br>933<br>043<br>8F4B | T3<br>T2PR<br>933<br>043<br>8F4B | T3<br>T3PR<br>931<br>023<br>12F4B | T6<br>TR<br>938<br>101<br>2F | T6<br>T1PR<br>933<br>043<br>8F4B | T6<br>T2PR<br>933<br>043<br>8F4B | | T7<br>TR<br>938<br>101<br>2F | T7<br>T1PR<br>933<br>043<br>8F4B | T7<br>T2PR<br>933<br>043<br>8F4B | T8<br>T2PR<br>933<br>043<br>8F4B |
| T9<br>T2PR<br>931<br>023<br>12F4B | T10<br>T2PR<br>933<br>043<br>8F4B | T11<br>T2PR<br>933<br>043<br>8F4B | T12<br>T2PR<br>933<br>043<br>8F4B | T13<br>TR<br>938<br>101<br>2F | T13<br>T1PR<br>933<br>043<br>2F | T13<br>T2PR<br>933<br>043<br>8F4B | T14<br>T2PR<br>933<br>043<br>8F4B | T15<br>T2PR<br>933<br>043<br>8F4B | T16<br>T2PR<br>933<br>043<br>8F4B | T16<br>T2PR<br>933<br>043<br>8F4B | T17<br>T3PR<br>931<br>023<br>12F4B | T18<br>T2PR<br>933<br>043<br>12F4B | T19<br>T2PR<br>933<br>043<br>8F4B |
| T21<br>T2PR<br>931<br>043<br>8F 4B | T22<br>T2PR<br>933<br>043<br>8F 4B | | T23<br>T2PR<br>933<br>043<br>8F 4B | T'A'<br>T2PR<br>931<br>023<br>12F 4B | | TT9 \| T1<br>T1<br>961<br>049<br>2x6F 2B | | | | | | | Test point |
| 12 \| 5<br>LCR \| LCR<br>961<br>049<br>2x6F 2B | 23 \| 20<br>LCR \| LCR<br>961<br>049<br>2x6F 2B | 34 \| 31<br>LCR \| LCR<br>961<br>049<br>2x6F 2B | 54 \| 44<br>LCR \| LCR<br>961<br>049<br>2x6F 2B | | | T4 \| T4<br>TJS \| TJR<br>962<br>1099<br>2x6F 2B | T14 \| T15<br>TJS \| TJR<br>962<br>1059<br>2x6F 2B | T15 \| T15<br>TJS \| TJR<br>862<br>1059<br>2x6F 2B | T16 \| T19<br>TJS \| TJR<br>962<br>1059<br>2x6F 2B | | Hooton RxR | Stanlow & Thornton R&R | Stanlow & Thornton R&R |
| | | | | | | | | | | | | | |
| | | | | | | | | | | | | | |

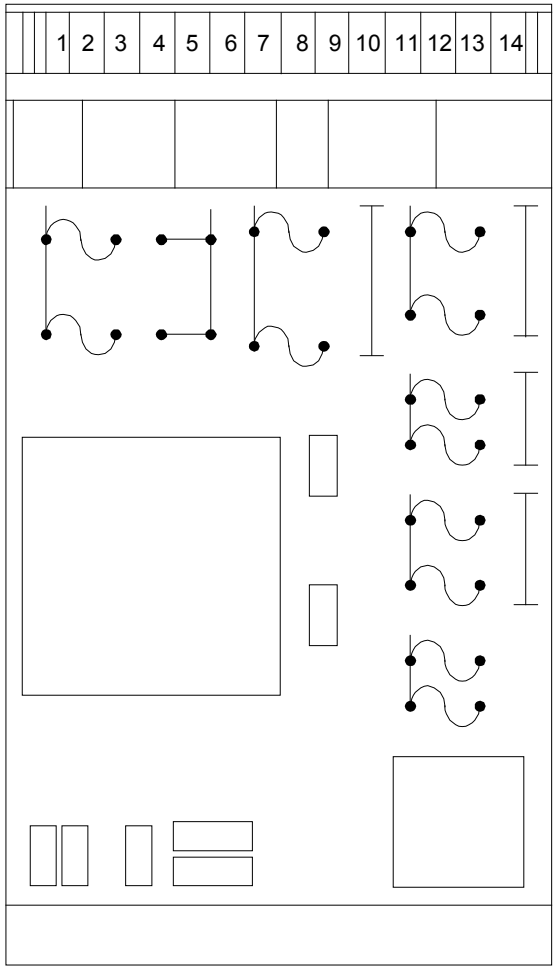
Figure 35

034

## 5.4 Location Case Profiles

The location case profiles detail the position of all equipment within **location cases**. Figure 36 provides an example of a location case profile. The description of the other equipment in the location case has been omitted for clarity.

*Typical Location Case Profile*




| No. | FUNCTION | TYPE | PINCODE | CONTACTS |
|---|---|---|---|---|
| 1 | 511HR | 930 | 004 | 8F 8B |
| 2 | 511HHR | 931 | 023 | 12F4B |
| 3 | 511(ST)UHR | 932 | 028 | 8F8B |
| 4 | 511DR | 931 | 023 | 12F4B |
| 5 | 511UECR | 942 | 070 | 2F2B |
| 6 | 511ECR | 941 | 071 | 4F |
| 7 | 511AYR | 934 | 063 | 8F4B |
| 8 | 505(FAIL)SR | 930 | 003 | 12F4B |
| 9L | SP | 961 | 049 | 6F2B |
| 9R | 852NWKPR | 961 | 049 | 6F2B |
| 10 | 513HR | 931 | 023 | 12F4B |
| 11 | 513DR | 931 | 023 | 12F4B |
| 12 | 513ECR | 941 | 071 | 4F |
| 13 | 511(4)UHR | 932 | 028 | 8F8B |
| 14 | 150TPR | 933 | 043 | 8F8B |

Note:Twin relay labelling. Top row=left side of relay.

Figure 36

035

## 5.5 Section Summary

This Section has looked at the

- relay room
- relay rack profile

Now complete Self-Assessment Question 5.

## Self-Assessment Question 5

State the information that is provided on a relay rack profile concerning each relay.

Now check your answer with the one given in the Answers Section. Make sure that you fully understand all the information given in Section Five. If there are any areas which you do not understand, re-read the relevant parts of the text.

# 6 Module Summary

This module has looked at

- the component parts which make up a relay
- the principles of operation of the relay
- the relay room and location cases

Before attempting the Self Test Questions which follow make sure that you fully understand what you have read. Take your time and check your answers to the Self-Assessment Questions again.

When you feel confident begin the Self Test Questions

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# 7 Self Test Questions

Attempt the following Self Test Questions. Take your time, and think about the answers carefully. When you have finished the Self Test check your answers with those given in the Answers Section. If you experienced any difficulties, return to the relevant part of the text and re-read the material.

Do not move on to the next module until you feel confident in your understanding of this work.

## Self Test Question 1

State the three principle components of the most common forms of relay.

## Self Test Question 2

Describe what happens to the front and back contacts when a non-latching relay is de-energised.

## Self Test Question 3

State the advantages of using a twin relay.

# Self Test Question 4

Describe how a thermal relay works.

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# 8 Answers

## Introduction

This section provides the answers to both the

- Self-Assessment Questions
- Self Test Questions

Check your answers carefully. If you experience any difficulties, re-read the relevant parts of the text.

Answers to Self-Assessment Questions

## Answer to Self-Assessment Question 1

Figure 37 illustrates the component parts of a relay correctly identified.

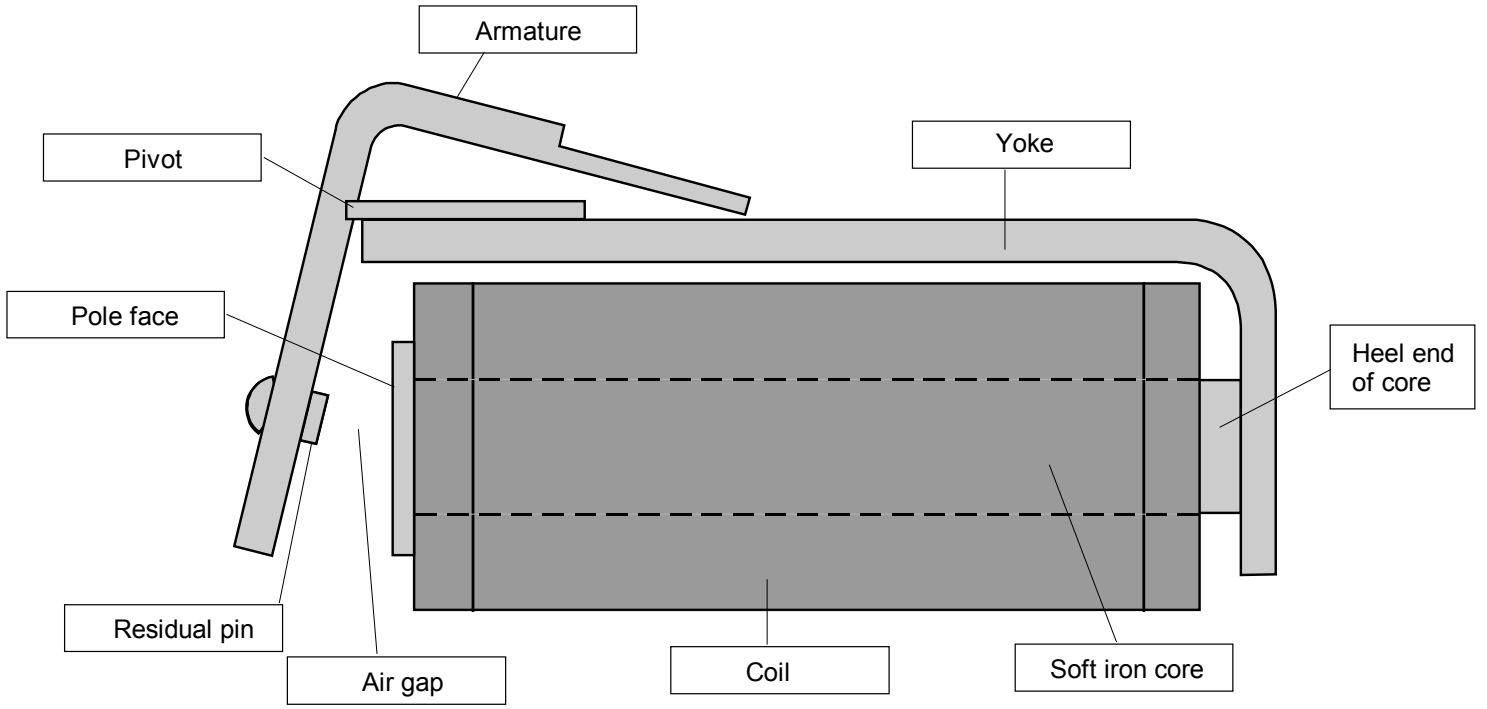

Armature
Pivot
Yoke
Pole face
Heel end of core
Residual pin
Air gap
Coil
Soft iron core
036

Figure 37

# Answer to Self-Assessment Question 2

In Figure 14

- current is flowing in the relay coil
- the magnetic field is beginning to build up
- the armature is being attracted towards the core

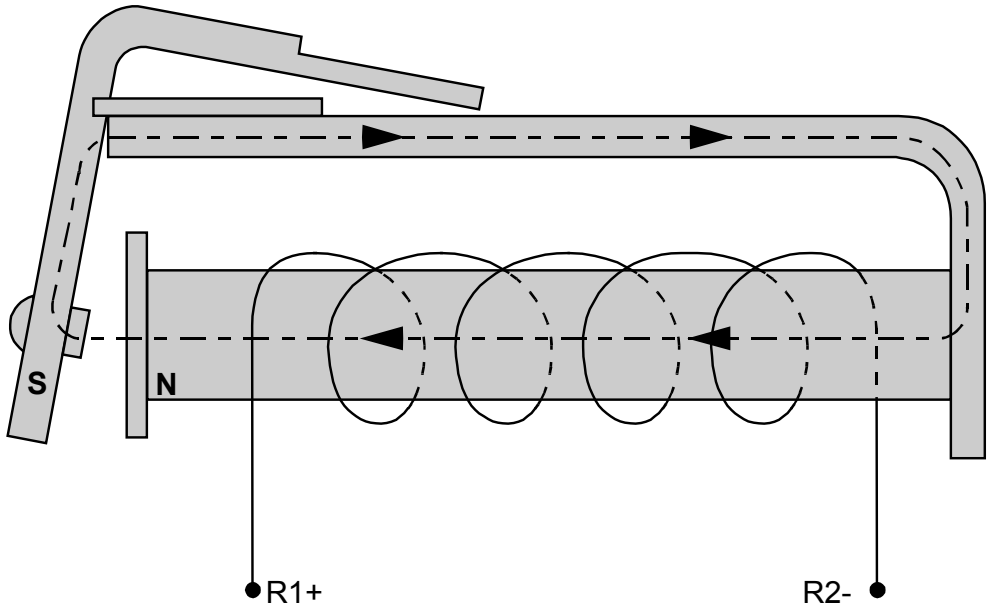



Figure 14

014

## Answer to Self-Assessment Question 3

When the relay is energised, contacts A1/A2 and A3/A4 are made.

When the relay is de-energised, contacts A5/A6 and A7/A8 are made.

## Answer to Self-Assessment Question 4

Figure 33 provides an example of a polar stick relay.

The polar stick relay is used where the operation of the relay

- is required to be dependent on the polarity of the current
- the contacts must remain in their last operated state when the operating current is removed

# Answer to Self-Assessment Question 5

The information that is provided on a relay rack profile is as follows

- relay reference
- BR Spec number
- pin code
- contact arrangement

# Answers to Self Test Questions

This section provides the answers to the Self Test Questions in Module Two. Compare them to your own answers. If you experience any difficulty re-read the relevant part of the text.

## Answer to Self Test Question 1

The three principle components of the most common forms of relay are

- coil
- armature
- contacts

## Answer to Self Test Question 2

When a relay is de-energised the

- front contacts are broken
- the back contacts are made

## Answer to Self-Test Question 3

The advantages of using a twin relay are

- reduced cost when compared to the purchase of two separate relays
- reduction of housing space required in the relay room
- reduced installation costs

## Answer to Self-Test Question 4

The thermal relay uses a bi-metallic strip which bends with heat and then cools down. Each time this occurs it closes a contact. The timing is complete once the cold contacts have made.

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# 9 Glossary

Within the railway system there are a number of regional variations and individual conventions which mean that alternative terms may be used for

- equipment
- operations
- features

Module Two has used terms which normally correspond with those found in The Signalling Installation Handbook. However, it may be useful for you to familiarise yourself with the alternative terminology which you may encounter in your work. This glossary provides a number of variations which may occur.

| | | |
|---|---|---|
| Circuit diagrams | - | Wiring diagrams<br>Drawings<br>Prints<br>Comics |
| Pole face | - | Core head |
| Yoke | - | Heel piece |
| Energise | - | Pick<br>Latch |
| Energised | - | Relay up<br>Relay picked<br>Latched |

| | | |
|---|---|---|
| De-energise | - | Drop<br>Release |
| De-energised | - | Relay down<br>Relay dropped<br>Released |
| BR 930 Series Relay | - | Q style |
| Plug board | - | Relay base |
| Beads | - | Wire labels |
| Slow to pick | - | Slow to pick up |
| Slow to release | - | Slow to drop<br>Slow to operate |
| Sleeve | - | Slug |
| Relay room | - | Equipment room<br>Walk in location<br>Relocatable equipment building (REB)<br>Location hut<br>Hut |
| Relay rack | - | Equipment rack |
| Location case | - | Location<br>Loc.(pronounced “lowk”)<br>Cupboard<br>Apparatus case<br>Location kiosk<br>Kiosk |